Ionic Equilibrium

DISCLAIMER

"The content provided herein are created and owned by various authors and licensed to Sorting Hat Technologies Private Limited ("Company"). The Company disclaims all rights and liabilities in relation to the content. The author of the content shall be solely responsible towards, without limitation, any claims, liabilities, damages or suits which may arise with respect to the same."

Ionic Equilibrium

Definitions

Electrolytes: Conduct electricity when dissolved in water (they produce ions).

(i) Strong Electrolytes:
Examples:.
HCl (aq.), HNO_3 (aq.), HClO (aq.), HI (aq.), NaCl (aq.), KNO_3 (aq.), HI (aq.), KNO_3 (aq.), NaOH (aq.), KOH (aq.) etc.

HBr (aq.) having concentration 'C' ionises as follows:

$$HBr\ (aq.) \longrightarrow H^+(aq.) + Br^-(aq.)$$
$$-\qquad\qquad C\qquad\qquad C$$

Similarly for KOH (aq.) solution;

$$KOH\ (aq.) \longrightarrow K^+(aq.) + OH^-(aq.)$$
$$-\qquad\qquad C\qquad\qquad C$$

Similarly for KNO_3 (aq.) solution;

$$KNO_3\ (aq.) \longrightarrow K^+(aq.) + NO_3^-(aq.)$$
$$-\qquad\qquad C\qquad\qquad C$$

Definitions

Strong electrolyte
Good conductors of electricity completely or almost completely ionized(means they break into ions) in water.

(ii) Weak Electrolytes:
Example:
Weak acids, Weak bases,

$HCN, CH_3COOH, C_6H_5COOH, NH_4OH, RNH_2$ etc.

NH_4OH (aq.) having concentration 'C' ionises as follows:

$$NH_4OH\ (aq.) \underset{\text{equilibrium}}{\rightleftharpoons} NH_4^+(aq.) + OH^-(aq.)$$
$$\text{At eqm. } C-x\qquad\qquad x\qquad\qquad x$$

Definitions

Weak electrolyte
- Ionises partially in water.
- Poor conductor of electricity.

(iii) Non-electrolytes:
Example: $C_6H_{12}O_6$ (aq.), $C_{12}H_{22}O_{11}$ (aq.) etc.

Definitions

Non-electrolyte
- Non ionized in water.
- Non-conductor of electricity.

Concepts of Acids and Bases

On the basis of litmus paper test:

Acids are those substances which turns **Blue litmus paper to Red.**

Bases are those substances which turns **Red litmus paper to Blue.**

Arrhenius Concept:

Acids:

Substances reacts with water and produce H_3O^+ are called acids.

Example: HCl(aq.), HBr(aq.), HNO_3(aq.) etc.

e.g

$$HNO_3(aq.) \longrightarrow H^+(aq.) + NO_3^-(aq.)$$

– C C

Bases:

Example: NaOH, KOH, NH_4OH etc.

$KOH\ (aq.) \longrightarrow K^+\ (aq.) + OH^-\ (aq.)$

$NH_4OH\ (aq.) \underset{\text{Indicates weak base}}{\rightleftharpoons} NH_4^+\ (aq.) + OH^-\ (aq.)$

Acid (aq.) + Base (aq.) → Salt (aq.) + Water (l)

$HCl\ (aq.) + NaOH\ (aq.) \rightarrow NaCl\ (aq.) + H_2O\ (l)$

$H^+\ (aq.) + Cl^-\ (aq.) + Na^+(aq.) + OH^-(aq.) \longrightarrow Na^+\ (aq.) + Cl^-(aq.) + H_2O\ (l)$

Net ionic equation: $H^+\ (aq.) + OH^-\ (aq.) \longrightarrow H_2O\ (l)$

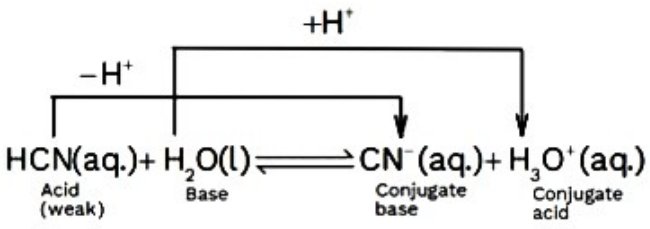
Bronsted-Lowry Concept:

Acid-base reaction is proton transfer reaction.

Acid: Proton donor(H^+).

Base: Proton acceptor(H^+).

HCN (aq.): Weak acid.



Note

Water is used as solvent.

Definitions

Acids:

Substances which furnishes H^+ ions in water.

Definitions

Base:

Substances which furnishes OH^- in water.

Points to remember

Arrhenius concept was based upon **ACID-BASE neutralisation**.

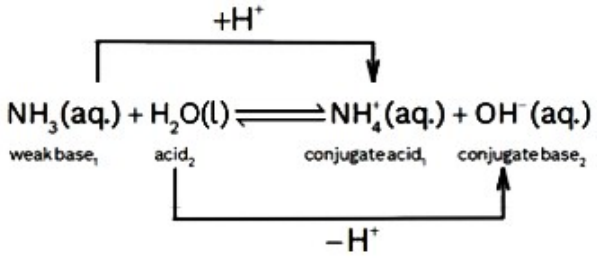
Points to remember

Na^+ and Cl^- ⟶ Spectator Ions (does not participate in reaction).

Note

- Any solvent is used within this concept.
- Concept based on formation of conjugate acid-base pair.

NH_3 (aq.): Weak base



Points to remember

H_2O (l) act as both acid and base.

Some important definitions

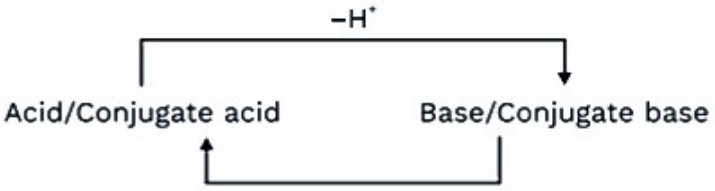
Strong acids: Acids stronger than H_3O^+ ion are called strong acids.
Example: HCl, HBr, HI etc.
$HBr\ (aq.) + H_2O\ (l) \rightarrow Br^-\ (aq.) + H_3O^+\ (aq.)$

Weak acids: Acids weaker than H_3O^+ ion are called weak acids.
Example: HCN, CH_3COOH etc.

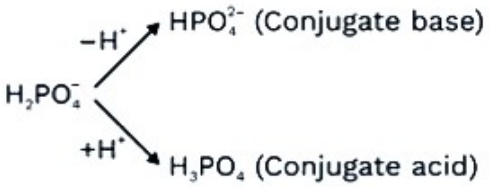
Strong bases: Base stronger than OH^- ion are called strong bases.
Example: $NH_2^- + H_2O \rightarrow NH_3\uparrow + OH^-$

Weak bases: Bases weaker than OH^- ion are called weak bases.
Example: $C_6H_5NH_2$, RNH_2 etc.

Amphiprotic Substances: Substance/ion which behaves as base and acid both.
Example: NaH_2PO_4, Na_2HPO_4, $NaHCO_3$ etc.



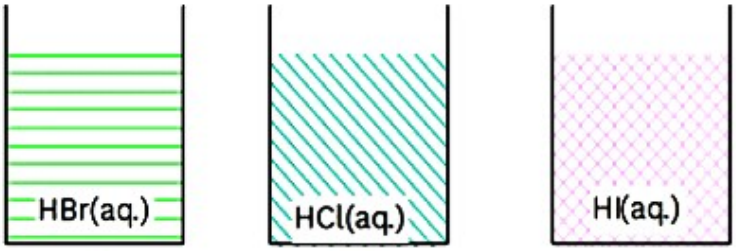
Position of equilibrium favoured forward formation of weak acid and weak base.



$$\underset{\text{acid}}{H_2PO_4^-(aq.)} + H_2O(l) \rightleftharpoons HPO_4^{2-}(aq.) + H_3O^+(aq.)$$

$$\underset{\text{base}}{H_2PO_4^-(aq.)} + H_2O(l) \rightleftharpoons H_3PO_4(aq.) + OH^-(aq.)$$

Leveling Effect of Water



0.1 M pH = 1 | 0.1 M pH = 1 | 0.1 M pH = 1

$HI\ (aq.) + H_2O\ (l) \rightarrow I^-(aq.) + H_3O^+(aq.)$

This effect is not shown by weak bases and acids because they ionise upto different extent in water.

Points to remember

All strong acids have same pH in water when above acids dissolve in water. They are equally ionised in water. They produce equal H^+ ions. This effect is known as **"Leveling Effect of Water."**



0.1 M | 0.1 M

Points to remember

Levelling effect of water is shown by strong bases and acids.

In ethanol (solvent):

Acid strength: $HClO_4 > HI > HBr > HCl > HNO_3$

Lewis Concept:

Coordinate covalent bond is formed during acceptance and donation of electron pair.

i.e. Acid + Base → Acid Base Adduct

$H^+(aq.) + H_2O(l) \longrightarrow H_3O^+(aq.)$

$H^+(aq.) + NH_3(l) \longrightarrow NH_4^+(aq.)$

$Ag^+(aq.) + 2NH_3(aq.) \longrightarrow Ag(NH_3)_2^+(aq.)$

Definitions

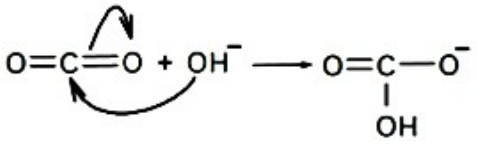
- Acids : Electron pair acceptor.
- Bases : Electron pair donor.

Cu^{+2} (aq.) + $4NH_3$ (aq.) $\longrightarrow Cu(NH_3)_4^{2+}$ (aq.)

BF_3 (aq.) + F^- (aq.) $\longrightarrow BF_4^-$ (aq.)

$SnCl_2$ (aq.) + $2Cl^-$ (aq.) $\longrightarrow SnCl_4^{2-}$ (aq.)

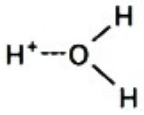
CO_2 (g) + OH^- (aq.) $\longrightarrow$ HCO_3^- (aq.)
Lewis acid Lewis base Acid-base adduct



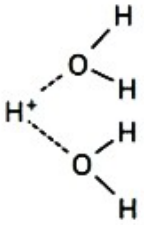
Existence of H^+ and OH^- in water

H^+ exists as $H^+(H_2O)_n$, where n = 1, 2, 3....

For n = 1: H_3O^+

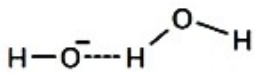


For n = 2: $H_5O_2^+$

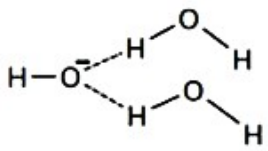


OH^- exists as $OH^-(H_2O)_n$,where n = 1,2,3....

For n = 1: $H_3O_2^-$



For n = 2: $H_5O_3^-$



Note

Bases have lone pair of electrons.
Example: F^-, Cl^-, CN^-, NH_3, H_2O etc.

Note

Acids:

Substance having

(i) Incomplete octet.
(ii) Vacant orbitals.
(iii) Atoms of different electronegative.

Ionisation constant and Ionic product of water:

Suppose $H_2O(l)$ having concentration 'C'.

$$\therefore \quad H_2O(l) \rightleftharpoons H^+(aq.) + OH^-(aq.)$$

At eqm. $C(1-\alpha) \quad C\alpha \quad C\alpha$

$$K_i = \frac{[H^+(aq.)][OH^-(aq.)]}{[H_2O(l)]}$$

$$K_i[H_2O(l)] = [H^+(aq.)][OH^-(aq.)]$$

K_d (equilibrium constant of dissociation) or $K_i \rightarrow$ Equilibrium constant of ionisation.

Or K_a = Acidic constant.

$$\boxed{K_w = [OH^-(aq.)][H^+(aq.)]}$$

where $K_w = K_i \cdot [H_2O(l)]$

K_w = Ionic product of water.

Or K_{SIC} : Self ionisation constant.

Or K_{APC} : Auto protolysis constant(or thermodynamic equilibrium constant).

$$H_2O(l) + H_2O(l) \rightleftharpoons H_3O^+(aq.) + OH^-(aq.)$$

$$K_w = [H_3O^+(aq.)][OH^-(aq.)]$$

$$NH_3(l) + NH_3(l) \rightleftharpoons NH_4^+(aq.) + NH_2^+(aq.)$$

$$K_{SIC} = [NH_4^+(aq.)][NH_2^+(aq.)]$$

Not $K_w \rightarrow$ Because its for water.

$$H_2O(l) \rightleftharpoons H^+(aq.) + OH^-(aq.)$$

$$C(C-\alpha) \quad C\alpha \quad C\alpha$$

At 25°C, $K_w = 10^{-14}$ (experimentally).

As $[H_2O(l)] = \frac{1000}{18}$ M

$$K_i = \frac{K_w}{[H_2O(l)]} = \frac{10^{-14}}{\frac{1000}{18}} = \boxed{1.8 \times 10^{-16}}$$

Note

K_w depends on temperature when temperature increases, K_w increases.

$$K_w = [H^+(aq.)][OH^-(aq.)] = C^2\alpha^2$$

$$\alpha = \frac{\sqrt{K_w}}{C} = \frac{10^{-7}}{1000/18}$$

∴ Degree of dissociation = 1.8×10^{-9}

$$\%\alpha = 1.8 \times 10^{-7}$$

- on T ↑, K_w ↑

$H_2O(l) \rightleftharpoons H^+(aq.) + OH^-(aq.)$ $\boxed{\Delta H = +ve}$

$H^+(aq.) + OH^-(aq.) \rightleftharpoons H_2O(l)$, $\boxed{\Delta H = -ve}$

From Van't Hoff equation we have:

$$\ln\left(\frac{K_2}{K_1}\right) = \frac{\Delta H}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$, where ΔH = J/mole , R = 8.314 J/mol.K

Put K_{w_1} = Ionic product at T_1; K_{w_2} = Ionic product at T_2

- pH_{water} ↓ when T↑

Temperature	K_w
11°C	0.31×10^{-14}
25°C	10^{-14}
50°C	5.5×10^{-14}
63°C	10^{-13}

pH Definition:

For aqueous dilute solution:

$$\boxed{pH = -\log[H^+(aq.)]}$$

Or $pH = -\log[H_3O^+(aq.)]$

(there must be no unit in log)

As per thermodynamics:-

$\boxed{pH = -\log a_{H^+}}$, where a_{H^+} is activity of H^+

As, $a_i = \gamma\frac{[i]}{C}$ but,

Note

pH scale was given by **S.Sorrenson.**

Note

Activity has no units.

For dilute solution: $\gamma = 1$

$\therefore \quad \boxed{a_i = \frac{[i]}{C^\circ}}$ where $C^\circ = 1$ M

$$a_{H^+} = \frac{[H^+]}{C^\circ}$$

$$pH = -\log\left(\frac{[H^+]}{C^\circ}\right) \text{ where } C^\circ = 1 \text{ M}$$

Points to remember

Similarly:-

$$\boxed{\begin{aligned} pOH &= -\log\frac{[OH^-]}{C^\circ} \\ pK_w &= -\log K_w \\ pK_a &= -\log K_a \end{aligned}}$$

pH Scale:

At 25°C, $K_w = 10^{-14}$ or $pK_w = 14$

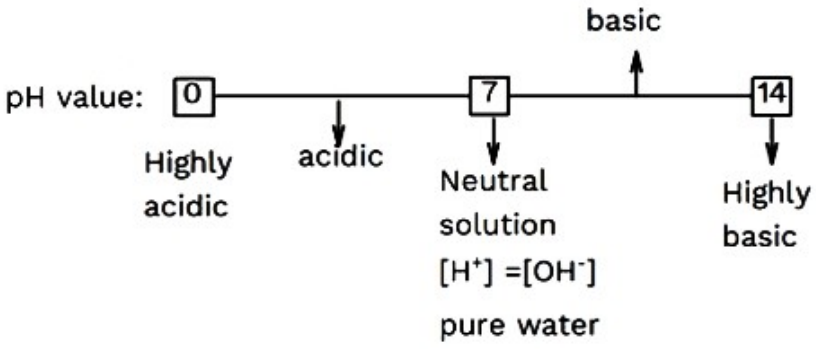
$$K_w = [H^+(aq.)][OH^-(aq.)]$$

$$-\log K_w = -\log[H^+] - \log[OH^-]$$

$$pK_w = pH + pOH$$

Points to remember

$$\boxed{14 = pK_w = pH + pOH}$$



(i) If $[H^+] > [OH^-]$ in solution then it is acidic solution and pH < 7 at 25°C.
(ii) If $[H^+] < [OH^-]$ in solution then it is basic solution and pH > 7 at 25°C.
(iii) If $[H^+] = [OH^-]$ in solution then it is neutral solution and pH = 7 at 25°C.

For neutral solution :- $[H^+] = [OH^-] = \sqrt{K_w}$ and K_w depends on temperature.

Points to remember

$[H^+]\uparrow$, $pH\downarrow$

At 25°C, $[H^+] = [OH^-] = 10^{-7}$ M

$pH = -\log[H^+]$

At 63°C,

0 ——————— 6.5 ——————— 13

Note

$\uparrow K_w = 10^{-13}$, $\downarrow pK_w = 13$.

∴ At 63°C
(i) pH < 6.5 of solution means it is acidic solution.
(ii) pH > 6.5 of solution means it is basic solution.
(iii) pH = 6.5 of solution means it is neutral solution.

At 11°C,
$K_w = 0.31 \times 10^{-14}$, $pK_w = 14.5$
∴ At 11° C
If pH < 7.25 of solution then it is acidic solution.
If pH > 7.25 of solution then it is basic solution.
If pH = 7.25 of solution then it is neutral solution.

Points to remember

In case pH = 7 of solution indicates acidic nature of solution which implies that pH = 7 doesn't represent a neutral solution always.

pH calculation of strong acid solutions:
Let we have a solution of HCl(aq.) having 'C' M concentration **at 25°C**.

Therefore, its decomposition can be elaborated as:

$$HCl(aq.) \rightleftharpoons H^+(aq.) + Cl^-(aq.)$$
$$_ \qquad C \qquad C$$

Let we have a 4 M HCl solution then,

$$pH = -\log\left[H^+(aq.)\right] = \boxed{-0.6}$$

Points to remember

But values of pH cannot be negative.

So, activity coefficient is required (γ)
If activity coefficient is not given, then consider pH of such solutions as nearly zero or solution is highly acidic.

HCl(aq.)	1 M	10^{-5} M	10^{-6} M	10^{-7} M	10^{-8} M	10^{-12} M
pH:	1	5	6	7	8	12
Weak acid	Correct	Correct	Correct (just less than 6)	×	×	×

$$\left[H^+\right]_{\text{Solution total}} = \left[H^+\right]_{HCl} + \left[H^+\right]_{H_2O}$$

In 10^{-7} M HCl :

$$[H^+]_{total} = [10^{-7} + 10^{-7}] = 2 \times 10^{-7}$$

$$pH = 6.7$$

(It is also not correct because violates Le-chatelier principle.)

$$HCl(aq.) \longrightarrow H^+(aq.) + Cl^-(aq.)$$

$$H_2O(l) \rightleftharpoons H^+(aq.) + OH^-(aq.)$$

At eqm.	-	10^{-7}	10^{-7}
At new cond.		$10^{-7} + 10^{-7}$	10^{-7}
At new eqm.		$(10^{-7} + x)$	x

where $\boxed{x < 10^{-7}}$

$$K_w = [H^+]_{total}[OH^-]$$

$$10^{-14} = (x + 10^{-7})(x)$$

$$\underset{\text{From } H_2O}{[H^+]} \quad \underset{\text{from HCl}}{[H^+]} \quad \underset{\text{from } H_2O}{[OH^-]}$$

$\boxed{x^2 + 10^{-7}x - 10^{-14} = 0}$ gives $\boxed{x = 0.618 \times 10^{-7}}$

$$[H^+]_{total} = 10^{-7} + 0.618 \times 10^{-7} = 1.618 \times 10^{-7}$$

$$\boxed{pH = 6.73}$$

In 10^{-6} M HCl(aq.):

$$K_w = [H^+]_{total}[OH^-]$$

$$10^{-14} = (10^{-6} + y)y \text{ where } y < 0.618 \times 10^{-7}$$

$$y \cong 10^{-8}$$

$$[H^+]_{total} = 10^{-6} + 10^{-8} \cong 10^{-6} \text{ M}$$

pH = 6 (or we can say just less than 6.)

Common Ion Effect

When we add HCl into water, due to common ion, $[H^+]$ ion equilibrium of water shifts in backward direction as per Le-chatelier's principle.
So, degree of ionisation of water decreases.
Let we have 10^{-8} M HCl(aq.) solution:

Points to remember

Because 10^{-6} M have more $[H^+]$ added therefore more backward reaction and also have more common ion effect.

Definitions

Degree of ionisation of weak electrolyte decreases due to addition of its one product-ion addition from other source is called common ion effect.

(It shows less common ion effect)

$10^{-14} = y(10^{-8} + y)$ where $y > 0.618 \times 10^{-7}$

$$y \cong 10^{-7}$$

$\therefore$ $[H^+] = 10^{-7} + 10^{-8} \cong 10^{-7}$

$\therefore$ pH = 7 (just less than 7)

Important: If concentration of H^+ from other source (like HCl) is less than 10^{-6} M (or 10^{-7} or less) then only we should consider concentration of H^+ ion from water as final concentration and

pH be like:

10^{-7} M HCl(aq.) have pH = 6.79

10^{-6} M HCl(aq.) have pH approximately equal to 6 (lesser than 6).

10^{-8} M HCl(aq.) have pH = 6.98

Concentration of $[H^+]$ from Strong Diprotic Acid and pH Calculation

Let we have strong diprotic acid be like H_2A (aq.) having concentration = 'C' M

It's dissociation can be elaborated as:

$$H_2A\,(aq.) \longrightarrow 2H^+\,(aq.) + A^{2-}\,(aq.)$$

$$-\qquad\qquad 2C\qquad\qquad C$$

Let we have 10^{-5} M H_2A(aq.) solution.

Then we have $[H^+] = 2 \times 10^{-5}$ M

Therefore, pH = 5 – log 2 = **4.7**

pH of Strong Base Solution

Let we have a solution of NaOH(aq.) having concentration = 'C' M

NaOH(aq.)	10^{-3}M,	10^{-6}M,	10^{-7}M,	10^{-8}M
pOH	3	6	6.75	6.98
pH	11	8	7.21	7.02

If concentration of OH^- from other source is less than 10^{-6} M, then we should consider same method to calculate concentration of OH^- from water.

Points to remember

Remember these Values

log 3 = 0.4771 ≅ 0.5

log 2 = 0.3010 ≅ 0.3

Note

For very dilute solution like below, we don't even approach to calculation part.

Note:

Let we have 10^{-12} M HCl(aq.) solution.

Then pH = 6.99 (Just less than 7) **at 25°C**

Note: pH calculation will be

Similarly, like strong acidic solutions.

Dissociation of NaOH (aq.) solution can be elaborated as:

$$NaOH(aq.) \longrightarrow Na^{+}(aq.) + OH^{-}(aq.)$$

$$-\qquad\qquad C\qquad\quad C$$

Here calculation is done as: $10^{-14} = [H^{+}][OH^{-}]_{total}$

pH of a resultant solution of a strong acid and strong base.

Let we have strong acid HCl(aq.) solution having molarity M_1,and volume = V_1
and strong base NaOH(aq.) solution having molarity M_2,and volume = V_2

$V_{sol.}$ (after mixing the two given solution) = $V_1 + V_2$

After mixing them :

Reaction : HCl (aq.) + NaOH (aq.) $\longrightarrow$ NaCl (aq.) + $H_2O(l)$

If millimoles of H^+ > millimoles of OH^- in the resultant solution, then solution is **acidic**	$[H^+]_{resultant} = \frac{\text{Millimoles of } H^+ - \text{Millimoles of } OH^-}{\text{Total volume}} = \frac{M_1V_1 - M_2V_2}{V_1 + V_2}$
If millimoles of H^+ < millimoles of OH^- in the resultant solution, then solution is basic.	$[OH^-]_{resultant} = \frac{M_2V_2 - M_1V_1}{V_1 + V_2}$
If **millimoles of H^+ = millimoles of OH^-** in the resultant solution,	then solution is neutral and pH = 7 of the resultant solution.

pH of a resultant solution of two strong acids :

Let us consider an acidic solution of HCl (aq.) having molarity M_1 and volume = V_1
Also, we have an acidic solution of HNO_3 (aq.) having molarity M_2 and volume = V_2
After mixing the above two solutions:

$$V_{resultant\ solution} = V_1 + V_2$$

$$[H^+]_{resultant} = \frac{\text{Total moles of } H^+}{\text{Total volume}} = \frac{M_1V_1 + M_2V_2}{V_1 + V_2}$$

If V_3 ml of $H_2O(l)$ is added then: $\left[H^+\right]_{resultant} = \frac{M_1V_1 + M_2V_2}{V_1 + V_2 + V_3}$

For a Weak-Conjugate Acid-Base Pair:

Case of Weak acid and its strong conjugate base:

Let we have an acidic solution of HA(aq.) which is a weak acid.

So, its conjugate base is A^-.
The dissociation of weak acid can be studied as:

$$HA\ (aq.) \rightleftharpoons H^+(aq.) + A^-(aq.)$$

$$\text{Or } HA(aq.) + H_2O(l) \rightleftharpoons H_3O^+(aq.) + A^-(aq.)$$

Ionisation constant of acid$(K_c/K_a) = \dfrac{[H_3O^+][A^-]}{[HA]}$ .....(1)

Let us assume one another solution of a salt which contains the conjugate part of the above acid i.e
NaA(aq.) having concentration 'C' M.
The dissociation of salt solution can be studied as below:

$$NaA(aq.) \rightarrow Na^+(aq.) + A^-(aq.)$$

$$C \qquad C$$

$$A^-(aq.) + H_2O(l) \rightleftharpoons HA(aq.) + OH^-(aq.)$$

K_h (hydrolysis constant) or $K_b = \dfrac{[HA][OH^-]}{[A^-]}$

(K_h term is used when ion react with water.)

From(i) and (ii)

$$K_{a[HA]} \cdot K_{b[A^-]} = [H_3O^+]\ [OH^-]$$

$$\boxed{K_a \cdot K_b = K_w}$$

Now take –log on both sides:

$$-\log K_{a(HA)} - \log K_{b(A^-)} = -\log K_w$$

Therefore: $\boxed{pK_a + pK_b = pK_w}$

Case of Weak Base and its strong conjugate acid:
Let we have a basic solution of RNH_2 (aq.) which is a weak base.
The equilibrium of above weak basic solution can studied as:

$$RNH_2(aq.) + H_2O(l) \rightleftharpoons RNH_3^+(aq.) + OH^-(aq.)$$

$$K_{b(RNH_2)} = \frac{[RNH_3^+][OH^-]}{[RNH_2]} \qquad ...(3)$$

Now dissociation of salt solution of RNH_3Cl can be studied as:

Note
Ions of strong acid/base do not react with water.

Note
At 25°C : $K_a K_b = 10^{-14}$

Points to remember

So, its conjugate acid is RNH_3^+.
(Source of RNH_3^+ is RNH_3Cl which is a salt.)

$$RNH_3Cl(aq.) \longrightarrow RNH_3^+(aq.) + Cl^-(aq.)$$

$$RNH_3^+(aq.) + H_2O(l) \rightleftharpoons RNH_2(aq.) + H_3O^+(aq.)$$

$$K_h \text{ or } K_{a[RNH_3^+]} = \frac{[RNH_2][H_3O^+]}{[RNH_3^+]} \quad ...(4)$$

From(3) and (4):

$$\boxed{K_{a[RNH_3^+]} K_{b[RNH_2]} = K_w = [H_3O^+][OH^-]}$$

pH of a weak acidic solution:

Let us consider an acidic solution of HA(aq.) having molarity 'C' M.
Its established equilibrium can be studied as follows:

$$HA(aq.) \rightleftharpoons H^+(aq.) + A^-(aq.)$$

At eqm.	$C(1-\alpha)$	$C\alpha$	$C\alpha$

$$K_{a[HA]} = \frac{[H^+][A^-]}{[HA]} = \frac{C\alpha^2}{1-\alpha}$$

If α is negligible w.r.t 1 ($\alpha < 0.05$)

As, $[H^+] = C\alpha = \boxed{[H^+] = \sqrt{K_a.C}}$

On taking –log on both sides of above equation:

$$-\log[H^+] = \frac{1}{2}(-\log K_a - \log C)$$

$$\boxed{pH = \frac{1}{2}(pK_a - \log C)}$$

*This formula is valid only when **($\alpha < 0.05$)**

Note

Then the above equation is reduced to: $K_a = C\alpha^2$

$$\boxed{\alpha = \sqrt{\frac{K_a}{C}}}$$

Note

where α : degree of ionisation or dissociation.
K_a: acidic constant.

NOTE: But if $K_a \cong C$, then from above relation $\alpha \cong 1$ which is incorrect approach. Then we need to solve quadratic equation.

$K_{a[HA]} = \frac{[H^+][A^-]}{[HA]} = \frac{C\alpha^2}{1-\alpha}$ to get the degree of dissociation and pH of solution.

Q1 **Calculate pH of 0.1 M HA (aq.) solution.(Given $K_a = 10^{-5}$)**

Sol. First of all check 'α'.

$$\alpha = \sqrt{\frac{10^{-5}}{10^{-1}}} = \boxed{0.01} \quad (\because \text{almost negligible})$$

$$\therefore \text{pH} = \frac{1}{2}\left(5 - \log 10^{-1}\right) = \frac{1}{2}(5+1) = \boxed{3}$$

(In exam first check whether α's value can be neglected or not.)

Q2 **Calculate pH of 0.01 M HA (aq.) solution. (Given $K_a = 10^{-5}$)**

Sol. First of all check 'α'.

$$\alpha = \sqrt{\frac{10^{-5}}{10^{-2}}} = \sqrt{10} \times 10^{-2} = 3.16 \times 10^{-2} = \boxed{0.0316}$$

$$\text{pH} = \frac{1}{2}\left(5 - \log(0.01)\right) = \frac{1}{2}(5+2) = \boxed{3.5}$$

Ostwald Dilution Law:

- Applicable on weak electrolytes i.e $\alpha \to$ negligible w.r.t 1, then

$$\boxed{\alpha \propto \frac{1}{\sqrt{C}}}$$

As, $\downarrow C = \frac{n_{solute}}{V_{solution} \uparrow}$ where n_{solute} is constant in dilution process.

If $\alpha \geq 0.05$ then solve quadratic equation for pH calculation.

Also we know that: $\boxed{\alpha \propto \sqrt{V}}$

$$\boxed{\begin{array}{l} C \downarrow\downarrow,\ V \uparrow\uparrow,\ \alpha \uparrow \\ * \text{ If } C \to 0,\ V \to \infty \text{ then } \alpha \to 1 \end{array}}$$

Because 'α' is negligible w.r.t. 1. Therefore, it could maximum be 1.

At infinite dilution, weak acid behaves as strong acid ($\alpha \to 1$) i.e $\boxed{100\% \text{ ionise}}$

Q3 **Calculate pH of 10^{-6} M CH_3COOH (aq.) solution. (Given $K_a = 1.8 \times 10^{-5}$)**

Sol. $\alpha = \sqrt{\frac{1.8 \times 10^{-5}}{10^{-6}}} = \sqrt{18} > 1.$

Hence, solve quadratic equation.

$$CH_3COOH(aq.) \rightleftharpoons CH_3COO^-(aq.) + H^+(aq.)$$

At eqm. $C(1-\alpha)$ $C\alpha$ $C\alpha$

$$K_a = \frac{C\alpha^2}{(1-\alpha)}$$

$$1.8 \times 10^{-5}(1-\alpha) - 10^{-6}(\alpha^2) = 0$$

On solving above quadratic equation:

$$\boxed{\alpha = 0.95}$$

Now $[H^+] = C\alpha = 0.95 \times 10^{-6}$

$$\boxed{pH = 6.022}$$

If we consider common ion effect and $[H^+]$ from water then $\boxed{pH = 6.022}$

pH calculation of weak basic solution:

Let us consider a weak basic solution of $NH_3(aq.)$ having concentration 'C' M.

Or $NH_4OH(aq.)$

$(K_b = 1.8 \times 10^{-5})$

Its dissociation theory will proceed as it is of weak acidic solution.

$NH_3(aq.) + H_2O(l) \rightleftharpoons NH_4^+(aq.) + OH^-(aq.)$

Or $NH_4OH(aq.) \rightleftharpoons NH_4^+(aq.) + OH^-(aq.)$

$C(1-\alpha)$ $C\alpha$ $C\alpha$

$$K_b = \frac{C\alpha^2}{(1-\alpha)}$$

If $\boxed{\alpha < 0.5}$ then $\boxed{K_b = C\alpha^2}$ $\boxed{\alpha = \sqrt{\frac{K_b}{C}}}$

pH of a resultant solution of weak acid and strong acid solution:

Let the strong acid: HA (aq.) having molarity = C_1

So,

$$\boxed{[OH^-] = C\alpha = C\sqrt{\frac{K_b}{C}}}$$

$$\boxed{pOH = \frac{1}{2}(pK_b - \log C)}$$

Note

If $\alpha \geq 0.05$ then solve quadratic equation and find α.

$$[OH^-] = C\alpha$$

If $K_b = C$, then also solve quadratic equation)

And weak acid: HB (aq.) having molarity = C_2

α = degree of dissociation HB in presence of HA.

α' = degree of dissociation of HB in absence of HA.

Common ion : H^+

If $\alpha' = \sqrt{\frac{K_a}{C_2}}$ is negligible then α is also negligible

($\alpha < \alpha'$) due to common ion effect.

$HA\ (aq.) \rightarrow H^+(aq.) + A^-(aq.)$ (Irreversible reaction)

$C_1 \qquad C_1$

$HB\ (aq.) \rightleftharpoons H^+\ (aq.) + B^-(aq.)$

$C_2(1-\alpha) \qquad (C_2\alpha + C_1) \qquad (C_2\alpha)$

$C_2 - x \qquad x + C_1 \qquad x$

$$K_a = \frac{[H^+]_{total}[B^-]}{[HB]} = \frac{(C_1 + x)(x)}{C_2 - x}$$

Solve quadratic equation to obtain pH.

Note

Where α' = degree of dissociation in absence of HA.

$\boxed{\text{where } x = C_2\alpha}$

Points to remember

If x is negligible w.r.t C_2: $\boxed{K_a = \frac{(C_1 + x)x}{C_2}}$

If strong acid solution is fairly concentrated, $\boxed{K_a = \frac{C_1}{C_2}x}$

(Or $C_1 = C_2$) Then, $\boxed{[H^+] = C_1 + x = C_1}$ $\boxed{[B^-] = x = C_2\alpha}$

Q4 **Calculate pH of mixture of 0.1 M HCl and 0.1 M HB. (K_a of HB = 10^{-5})**

Sol. $\alpha' = \sqrt{\frac{10^{-5}}{0.1}} = \boxed{0.01}$

$C_1 = C_2 = 0.1M,\ x = \frac{C_2}{C_1}K_a = \boxed{10^{-5}} = [B^-]$

$\therefore\quad [H^+]_{total} = C_1 + x = 0.1 + 10^{-5} = \boxed{0.1}$

$pH = 1$

To check: Common ion effect: $x = C_2\alpha$ $\qquad \boxed{\alpha = 10^{-4}}$

pH of resultant solution of two weak acids:

Let us consider two weak acidic solution be HA(aq.) have credentials K_{a_1}, C_1, α_1 and HB(aq.) having credentials K_{a_2}, C_2, α_2.

α_1 and α_2 → D.O.I. in presence of each other.
α'_1 and α'_2 → D.O.I. in absence of each other.
Clearly, $\alpha_1 < \alpha_1'$ and $\alpha_2 < \alpha_2'$ due to common ion effect.

$$\alpha'_1 = \sqrt{\frac{K_{a_1}}{C_1}};\ \alpha_2^1 = \sqrt{\frac{K_{a_2}}{C_2}}$$

	$HA(aq.)$	$\rightleftharpoons$ $H^+(aq.)$	$+$ $A^-(aq.)$	
At eqm	$C_1(1-\alpha_1)$	$(C_1\alpha + C_2\alpha_2)$	$C_1\alpha$	
	$C_1 - x$	$(x+y)$	x	where $\boxed{x = C_1\alpha_1}$

	$HB(aq.)$	$\rightleftharpoons$ $H^+(aq.)$	$+$ $B^-(aq.)$	
At eqm	$C_2(1-\alpha_2)$	$(C_2\alpha_2 + C_1\alpha_1)$	$C_2\alpha_2$	
	$C_2 - y$	$(x+y)$	y	where $\boxed{y = C_2\alpha_2}$

$$K_{a_1} = \frac{[H^+]_{total}[A^-]}{[HA]} = \frac{(x+y)x}{C_1 - x} \qquad ...(1)$$

$$K_{a_2} = \frac{[H^+]_{total}[B^-]}{[HB]} = \frac{(x+y)y}{C_2 - y} \qquad ...(2)$$

Assumption:

If x is negligible w.r.t. is C_1 and y is negligible w.r.t. is C_2

$$(x + y)x = K_{a_1}C_1 \quad \ldots(3)$$

$$(y + x)y = K_{a_2}C_2 \quad \ldots(4)$$

Add $(x+y)^2 = K_{a_1}C_1 + K_{a_2}C_2$

Or $\left[H^+\right]^2 = K_{a_1}C_1 + K_{a_2}C_2$

$\therefore \quad \left[H^+\right] = \sqrt{K_{a_1}C_1 + K_{a_2}C_2}$

Divide (3) and (4): $\frac{x}{y} = \frac{K_{b_1}C_1}{K_{b_2}C_2}$

Solve for x and y.

After that check if your assumption was right or not.

pH of resultant solution of two weak bases:

Let: XOH (aq.): K_{b_1}, C_1, α_1 be one of the weak basic solution

And YOH (aq.): K_{b_2}, C_2, α_2 be another weak basic solution.

If x and y are negligible respectively, then

$$\left[OH^-\right] = x + y = \sqrt{K_{b_1}C_1 + K_{b_2}C_2}$$

$$\frac{\left[X^+\right]}{\left[Y^+\right]} = \frac{x}{y} = \frac{K_{b_1}C_1}{K_{b_2}C_2}$$

Note

All steps are similar as above in derivation within this case.

Q5 **Calculate pOH of 10^{-4} M HA solution which is 4% ionised.**

Sol. pOH = ?

$[HA] = 10^{-4}$

$\alpha = 4\%$

	HA	$\rightleftharpoons$	H^+	+	A^-
t = 0	C		0		0
$t = t_{eq}$	C(1 – α)		Cα		Cα
	[H+]	=	Cα	=	$10^{-4} \times 4/100 = 4 \times 10^{-6}$

$pH = -\log [H^+] = -\log (4 \times 10^{-6})$

$= 6 - 2\log 2$

$= 6 - 0.60\ 4$

$\cong 5.4$

$pHO = 14 - pH = 14 - 5.4 = 8.6$

Q6 **If α is 0.01 for centimolar solution (M/100) of weak acid (HA), then find the value of $pK_{b(A^-)}$ at 25°C.**

Sol. $HA\ (aq.) \rightleftharpoons H^+\ (aq.) + A^-(aq.)$

$C(1-\alpha) \quad C\alpha \quad C\alpha$

$K_a = \frac{C\alpha^2}{1-\alpha}$, given $\alpha = 0.01$

$K_a = C\alpha^2 = \frac{1}{100} \times 10^{-4} = 10^{-6}$

$pK_a = 6$

As, $pK_a + pK_b = pk_w$

$pK_b = 14 - 6 = 8$

Q7 **Calculate molar concentration of NH_4OH solution that has pH = 12 ($K_b = 1.8 \times 10^{-5}$)**

Sol. $NH_4OH \rightleftharpoons NH_4^+(aq.) + OH^-(aq.)$

$C(1-\alpha) \quad C\alpha \quad C\alpha$

pOH = 2

$\therefore [OH^-] = C\alpha = 10^{-2}$

$K_b = \frac{C\alpha^2}{1-\alpha}$ (neglect α w.r.t 1)

$1.8 \times 10^{-5} = C\alpha^2 = 10^{-2} \times \alpha$

$\alpha = 1.8 \times 10^{-3}$

$\therefore\ C = \dfrac{10^{-2} \times 10^{+3}}{1.8} = \dfrac{10}{1.8} = 5.55$

Q8 **How much water must be added to 10 L of 0.2 M of HA for DOI for to be double?**
($K_a = 1.8 \times 10^{-6}$)

Sol.

$$HA \rightleftharpoons H^+ + A^-$$

$0.2(1-\alpha) \quad 0.2\alpha \quad 0.2\alpha$

$$\alpha = \sqrt{\frac{1.8 \times 10^{-6}}{0.2}} = \boxed{3 \times 10^{-3}}$$

$$\therefore \quad 2\alpha = 6 \times 10^{-3} = \sqrt{\frac{1.8 \times 10^{-6}}{C}}$$

$1/C = 20$

$$20 = \frac{V}{2}$$

$\therefore\ V = 40$

Therefore, 30 L water must be added.

pH calculation of Polyprotic Acid Solution:

Let us consider diprotic acid be H_2A type: ($H_2C_2O_4$, H_2CO_3, H_2S)

K_{a_1}: First ionisation constant.

K_{a_2}: Second ionisation constant.

where $\boxed{K_{a_1} > K_{a_2}}$

'C': Initial concentration of weak diprotic acid.

$$H_2A(aq.) \rightleftharpoons H^+(aq.) + HA^-(aq.)$$

At equilibrium $\quad C-x \quad \boxed{x+y} \quad \boxed{x-y}$

$$HA^-(aq.) \rightleftharpoons H^+(aq.) + A^{2-}(aq.)$$

At equilibrium $\quad \boxed{x-y} \quad \boxed{x+y} \quad y$

C–x = concentration of H_2A when both equilibrium simultaneously established.

y = concentration of A^{2-} when both equilibrium simultaneously established.

$$K_{a_1} = \frac{[H^+]_{total}[HA^-]}{[H_2A]} = \frac{(x+y)(x-y)}{(C-x)}$$

$$K_{a_2} = \frac{[H^+]_{total}[A^{2-}]}{[HA^-]} = \frac{(x+y)y}{x-y}$$

Assumption (1):

If there is large difference between K_{a_1} and K_{a_2} ($K_{a_1} >> K_{a_2}$), then neglect 'y' w.r.t 'x'.

$$\boxed{K_{a_1} = \frac{x^2}{C-x}} \qquad \boxed{K_{a_2} = y}$$

Assumption (2):

If 'x' is negligible w.r.t 'C'.

$$\boxed{x = \sqrt{K_{a_1}.C}}$$

Then:

$$[H^+]_{total} = x + y \simeq x$$

$$[HA^-] = x - y \simeq x$$

$$[A^{2-}] = y$$

Q9 **Calculate pH of 0.01 M H_2A (aq.) solution.**
(Given: $K_{a_1} = 10^{-6}$ and $K_{a_2} = 10^{-9}$)

Sol. $x = \sqrt{K_{a_1}.C} = 10^{-4}$

$y = K_{a_2} = 10^{-9}$

$[H^+] = x + y = x = 10^{-4}$ if 'y' is negligible w.r.t. 'x'

$\boxed{pH = 4}$

Case -II:

Q10 **If HBr is added in solution and pH of resultant solution is 3 then calculate concentration of A^{2-}[COMMON ION EFFECT]**

Sol. $H_2A(aq.) \rightleftharpoons 2H^+(aq.) + A^{2-}(aq.)$

Adding both reactions of simultaneous equilibria:

$$K_{a_1}.K_{a_2} = \frac{\left[H^+\right]^2_{total}\left[A^{2-}\right]}{\left[H_2A\right]}$$

$$10^{-6} \times 10^{-9} = \frac{\left(10^{-3}\right)^2.\left[A^{2-}\right]}{0.01} \quad (x' << 0.5)$$

$$[A^{2-}] = 10^{-11} < y$$

Due to Common Ion Effect.

H_2A	$\rightleftharpoons$	H^+	+	HA^-
C-x'		x' + y' + z		x' - y'
HA^-	$\rightleftharpoons$	H^+	+	A^{2-}
x' –y'		x' + y' + z		y'

pH calculation of weak triprotic acid like H_3PO_4(aq.) solution:

H_3PO_4 is a weak triprotic acid, having:

$$K_{a_1} = 7 \times 10^{-3}, \quad K_{a_2} = 6 \times 10^{-8}, \; K_{a_3} = 5 \times 10^{-13}$$

Let 'C': Initial concentration of H_3PO_4

Its dissociation can be studied as follows:

	H_3PO_4 (aq.)	$\rightleftharpoons$	H^+ +	$H_2PO_4^-$
At equilibrium	C-x		(x+y+z)	(x-y)
	$H_2PO_4^-$ (aq.)	$\rightleftharpoons$	H^+ +	HPO_4^{2-}
At equilibrium	(x – y)		(x+y+z)	(y-z)
	HPO_4^{-2}(aq.)	$\rightleftharpoons$	H^+ +	PO_4^{3-}
At equilibrium	(y – z)		(x+y+z)	z

z = concentration of PO_4^{3-} when all equilibrium are simultaneously established.

$$K_{a_1} = \frac{[H^+][H_2PO_4^-]}{[H_3PO_4]} = \frac{(x+y+z)(x-y)}{(C-x)}$$

$$K_{a_2} = \frac{[H^+][HPO_4^{2-}]}{[H_2PO_4^-]} = \frac{(x+y+z)(y-z)}{(x-y)}$$

$$K_{a_3} = \frac{[H^+][PO_4^{3-}]}{[HPO_4^{2-}]} = \frac{(x+y+z)(z)}{(y-z)}$$

Due to large difference between K_{a_1}, K_{a_2}, K_{a_3}

$$K_{a_1} >> K_{a_2} >> K_{a_3}$$

We can neglect 'y' w.r.t 'x' and 'z' w.r.t 'y'.

$$\boxed{K_{a_1} = \frac{x^2}{C-x}} \quad \boxed{K_{a_2} = y} \quad \boxed{K_{a_3} = \frac{xz}{y}}$$

$$\boxed{K_{a_1} = 7 \times 10^{-3}}$$

Can't take assumption that 'x' is negligible w.r.t 'C' because $K_{a_1} = 7 \times 10^{-3}$ and concentration is also very low.

$\therefore \alpha$ can't be neglected.

$$[H^+] = x + y + z = x$$

$$[HPO_4^{2-}] = y - z = y$$

$$[PO_4^{3-}] = z$$

pH calculation of weak diacidic base:

Let $\begin{matrix} CH_2 - NH_2 \\ | \\ CH_2 - NH_2 \end{matrix}$ $\begin{matrix} K_{b_1} \text{ and } K_{b_2} \\ \\ (\text{where } K_{b_1} >> K_{b_2}) \end{matrix}$ be weak diacidic base and given are ionisation constants.

$$CH_2-NH_2 \;|\; CH_2-NH_2 \quad + \quad H_2O(l) \rightleftharpoons CH_2-NH_3^+ \;|\; CH_2-NH_2 \quad + \quad OH^-$$

At equilibrium	C − x	−	(x − y)	(x + y)

$$CH_2-NH_3^+ \;|\; CH_2-NH_2 \quad + \quad H_2O(l) \rightleftharpoons CH_2-NH_3^+ \;|\; CH_2-NH_3^+ \quad + \quad OH^-$$

At equilibrium	(x-y)	−	y	y + x

Therefore $K_{b_1} = \dfrac{(x-y)(x+y)}{C-x}$ $\qquad$ $K_{b_2} = \dfrac{(x+y)y}{x-y}$

Assumption:

If $K_{b_1} >> K_{b_2}$ then neglect 'y' w.r.t 'x'.

$$\boxed{K_{b_1} = \frac{x^2}{C-x}}, \quad \boxed{K_{b_2} = y}$$

$$[OH^-] = x + y \simeq x$$

Take approximation.

Definitions

When cation/anion/ or both ions of salt reacts with water then process is called salt hydrolysis.

Salt Hydrolysis

Types of salt:

(1) **Complete Salt** :Formed by complete neutralisation of acid and base.

(a) Salts of monobasic acid-base. e.g: NaCl, CH_3COONa, NH_4Cl, CH_3COONH_4, NaH_2PO_2 etc.

(b) Salts in which acid or base is poly:

Poly acid and mono base: (i) $Na_2C_2O_4$, Na_2CO_3, Na_2S, Na_2HPO, Na_2PO_4

Poly base and mono acid: (ii) $Zn(NO_3)_2$, $MgCl_2$, $FeCl_2$

(2) **Incomplete Salts:** Formed by incomplete neutralisation of acid and base.

(a) **Acidic salts**: In which more base can be added.

$NaHPO_4$, Na_2HPO_4, $Na_2H_2PO_3$, $NaHCO_3$, NaHS, etc.

(b) **Basic salts**: In which more acid can be added.

Mg(OH)Cl, $Zn(OH)NO_3$, etc.

(A) **Salts of monobasic acid and monoacidic base.**

(I) Salts of strong acid and strong base:	NaCl, KCl, KNO_3, $NaClO_4$, KBr etc.
(II) Salts of strong base and weak acid: CH_3COONa, C_6H_5COONa, $HCOONa$, KNO_2,NaCN, etc.	$pH = \frac{1}{2}(pK_w + pK_a + \log C)$
(III) Salts of strong acid and weak base: NH_4Cl, NH_4NO_3, $C_6H_5NH_3Cl$, RNH_3NO_3, etc.	$pH = \frac{1}{2}(pK_w - pK_b - \log C)$
(IV) Salts of weak acid and weak base: CH_3COONH_4, RNH_3CN, NH_4CN, etc.	$pH = \frac{1}{2}(pK_w - pK_b + pK_a)$

Points to remember

All dissolved salts in water are strong electrolytes.

(I) **pH calculation of salt of strong acid and strong base:**

Let we have a salt solution of NaCl(aq.) having concentration 'C' M.

Its dissociation can be elaborated as:

$$NaCl(aq.) \longrightarrow Na^+ + Cl^-(aq.)$$

$$-\qquad\qquad C \qquad C$$

No hydrolysis takes place in such cases.

$[H^+] = [OH^-]$ means Neutral solution and pH = $\sqrt{K_w}$ = 7 at 25°C.

Points to remember

Na^+ and Cl^- do not react with H_2O. Cation of strong base and anion of strong acid do not react with water. Solution remains neutral.

(II) **pH calculation of salt solution of weak acid and strong base**

Let's we have a salt solution of CH_3COONa(aq.) having concentration 'C' M

$$CH_3COONa(aq.) \longrightarrow CH_3COO^- + Na^+$$

$$-\qquad\qquad C \qquad\qquad C$$

React with water ↓

Bronsted base

Points to remember

Anionic hydrolysis takes places in such cases.

Solution is always basic.

Note

$$\boxed{\%\ D.O.H = \propto \times 100}$$

pH > 7 at 25°C.

$CH_3COO^-(aq.) + H_2O(l) \rightleftharpoons CH_3COOH(aq.) + OH^-$

$C(1-\alpha)$ – $C\alpha$ $C\alpha$

Note

For a weak conjugate acid-base pair: $\boxed{K_a.K_b = K_w}$

$$K_{b(CH_3COO^-)} \text{ or hydrolysis constant } (K_h) = \frac{K_w}{K_{a(CH_3COOH)}} = \frac{C\alpha^2}{1-\alpha}$$

Alternate way to get K_h:

$$CH_3COO^- + H^+ \rightleftharpoons CH_3COOH \quad ; \quad \left(\frac{1}{K_a}\right)$$

$$+ \quad H_2O(l) \rightleftharpoons H^+ + OH^- \quad ; \quad K_w$$

$$CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$$

$$\therefore \quad K_b = K_w \times \frac{1}{K_a} = \frac{K_w}{K_a}$$

K_w : Water dissociation constant.

α : degree of hydrolysis

If α is not negligible w.r.t. 1 ($\alpha \geq 0.05$) then solve quadratic and find α.

NOTE: If $K_h \simeq C$, solve quadratic equation.

If α is negligible w.r.t. 1, then

$$K_h = \frac{K_w}{K_a} = C\alpha^2$$

$$\left[OH^-\right] = C\alpha = \sqrt{\frac{K_w.C}{K_a}}$$

$$\left[H^+\right] = \frac{K_w}{\left[OH^-\right]} = \frac{K_w}{\sqrt{K_w}}\sqrt{\frac{K_a}{C}} = \sqrt{\frac{K_w.K_a}{C}}$$

$$-\log\left[H^+\right] = \frac{1}{2}(-\log K_w - \log K_a + \log C)$$

pH calculation of salt solution of strong acid and weak base

Let us consider a salt solution of $NH_4Cl(aq.)$ having concentration 'C' M.

Its dissociation can be elaborated as:

$NH_4Cl(aq.) \rightleftharpoons NH_4^+(aq.) + Cl^-(aq.)$

– C C

Points to remember

$$\alpha = \sqrt{\frac{K_h}{C}} = \sqrt{\frac{K_{b(CH_3COO^-)}}{C}} = \sqrt{\frac{K_w}{K_aC}}$$

Note

$$\boxed{pH = \frac{1}{2}(pK_w + pK_a + \log C)}$$

where C = concentration anion/ salt; same when monoprotic salt.

NH_4^+ (aq.) + $H_2O(l) \rightleftharpoons NH_4OH + H^+$

Or, NH_4^+(aq.) + $H_2O(l) \rightleftharpoons NH_3 + H_3O^+$

$C(1-\alpha)$ – $C\alpha$ $C\alpha$

$$K_{a(NH_4^+)} \text{ or } K_h = \frac{K_w}{K_{b(NH_3)}} = \frac{C\alpha^2}{(1-\alpha)}$$

$$K_{a(NH_4^+)} . K_{b(NH_3)} = K_w$$

Alternatively:

$$NH_4^+ + OH^- \rightleftharpoons NH_3 + H_2O(l); \quad \frac{1}{K_b}$$

$$+ \quad 2H_2O \rightleftharpoons H_3O^+ + OH^-; \quad K_w$$

$$NH_4^+ + H_2O(l) \rightleftharpoons NH_3 + H_3O^+; \quad K_h = \frac{K_w}{K_b}$$

If α is not negligible w.r.t. 1 ($\alpha \geq 0.05$), solve quadratic equation. and $[H_3O]^+ = C\alpha$ then find pH.

If $\alpha < 0.5$, $K_h = \frac{K_w}{K_b} = C\alpha^2$ $\alpha = \sqrt{\frac{K_h}{C}} = \sqrt{\frac{K_w}{K_b . C}}$

$$\left[H_3O^+\right] = C\alpha = \sqrt{\frac{K_w . C}{K_b}}$$

$$-\log\left[H_3O^+\right] = \frac{1}{2}(-\log K_w - \log C + \log K_b)$$

Points to remember

- Cationic hydrolysis takes place in such cases.
- Solution is always acidic.
- pH < 7 at 25°C.

Note

$$K_h = \frac{K_w}{K_b}$$

Points to remember

$$pH = \frac{1}{2}(pK_w - pK_b - \log C)$$

where C = concentration of cation/ salt.

pH Calculation of Salt Solution of Weak Acid and Weak Base:

Let we have a salt solution of CH_3COONH_4(aq.) having concentration 'C' M.

Its dissociation can be elaborated as:

$CH_3COONH_4 \rightarrow CH_3COO^-$(aq.) + NH_4^+(aq.)

– C C

CH_3COO^-(aq.) + NH_4^+(aq.) + $H_2O(l) \rightleftharpoons CH_3COOH$(aq.) + NH_4OH(aq.)

$C(1-\alpha)$ $C(1-\alpha)$ $C\alpha$ $C\alpha$

Note

Cationic and anionic hydrolysis take place in such cases.

Solution may be acidic, basic or neutral depending upon K_b and K_a.

$$K_h = \frac{K_w}{K_a.K_b} = \frac{C^2\alpha^2}{C^2(1-\alpha)^2} = \frac{\alpha^2}{(1-\alpha)^2}$$

$$\sqrt{K_h} = \sqrt{\frac{K_w}{K_aK_b}} = \frac{\alpha}{1-\alpha}$$

Derivation:

(i) $CH_3COO^-(aq.) + H^+(aq.) \rightleftharpoons CH_3COOH \rightarrow 1/K_a$

(ii) $NH_4^+(aq.) + OH^-(aq.) \rightleftharpoons NH_4OH \rightarrow 1/K_b$

(iii) $H_2O(l) \rightleftharpoons H^+ + OH^- \rightarrow K_w$

On adding above three equation we get:

$CH_3COO^-(aq.) + NH_4^+(aq.) + H_2O(l) \rightleftharpoons CH_3COOH + NH_4OH$

Use (i) to get $\rightarrow [H^+]$

Use (ii) to get $\rightarrow [OH^-]$

(i) $CH_3COO^-(aq.) + H^+(aq.) \rightleftharpoons CH_3COOH(aq.) : 1/K_a$

$C(1-\alpha)$? $C\alpha$

$$\frac{1}{K_a} = \frac{C\alpha}{C(1-\alpha)[H^+]}$$

$$[H^+] = \frac{\alpha}{(1-\alpha)}K_a = \sqrt{\frac{K_w}{K_aK_b}}K_a = \boxed{\sqrt{\frac{K_wK_a}{K_b}}}$$

Points to remember

$$K_h = \frac{K_w}{K_aK_b}$$

Points to remember

$$pH = \frac{1}{2}(pK_w + pK_a - pK_b)$$

This equation does not depends on 'C'.

End Result

If $K_a > K_b$, $pK_a < pK_b$ therefore solution will be acidic and pH < 7.

If $K_a = K_b$, $pK_a = pK_b$ therefore solution will be neutral and pH = 7.

If $K_a < K_b$, $pK_a > pK_b$ therefore solution will be basic and pH > 7.

For $pK_a = pK_b$, example $K_{a(CH_3COOH)} = K_{b(NH_4OH)} = 1.8 \times 10^{-5}$

$$pH = 7$$

Assumption: Degree of hydrolysis is same for both ions.

When cation and anion hydrolyse in presence of each other their degree of hydrolysis is increases.

Hydrolysis of Polyvalent Ions

Let us consider a salt solution of type $Na_2A(aq.)$ having concentration 'C' M.

Its dissociation can be elaborated as:

$$Na_2A(aq.) \longrightarrow 2Na^+(aq.) + A^{2-}(aq.)$$

$$- \qquad 2C \qquad C$$

(Bronsted base)

$$A^{2-}(aq.) + H_2O(l) \rightleftharpoons HA^-(aq.) + OH^-(aq.)$$

At equilibrium: $(C-x) \quad - \quad (x-y) \quad (x+y)$

$$HA^- + H_2O(l) \rightleftharpoons H_2A + OH^-$$

$$(x-y) \quad - \quad y \quad (x+y)$$

$$K_{b(A^{2-})} \text{ or } K_{h_1} = \frac{K_w}{K_{a_2}} \qquad \text{[from (1)]}$$

$$K_{b(HA^-)} \text{ or } K_{h_2} = \frac{K_w}{K_{a_1}} \qquad \text{[from (2)]}$$

where $K_{h_1} > K_{h_2}$ because A^{2-} has more H^+ gain tendency.

$$K_{b(A^{2-})} . K_{a(HA^-)} = K_w$$

$\boxed{K_{h_1} . K_{a_2} = K_w}$ From (1)

$$K_{b(HA^-)} . K_{a(H_2A)} = K_w$$

$\boxed{K_{h_2} . K_{a_1} = K_w}$ From (2)

$$K_{h_1} = \frac{K_w}{K_{a_2}} = \frac{(x-y)(x+y)}{(C-x)}$$

$$K_{h_2} = \frac{K_w}{K_{a_1}} = \frac{y(x+y)}{(x-y)}$$

If $K_{h_1} >> K_{h_2}$ then neglect 'y' w.r.t 'x'.

If 'x' is negligible w.r.t 'C' then neglect. 'x' w.r.t 'C'.

Then we get: $\boxed{K_{h_2} = y}$ $\quad x = \sqrt{K_{h_1}.C}$

$[OH^-] = x + y \simeq x$

$[HA^-] = x - y \simeq x$

$[H_2A] = y$

Hydrolysis of Na_3PO_4 Salt Solution:

Let us consider a salt solution of Na_3PO_4 (aq.) having concentration 'C' M.

$$Na_3PO_4 \longrightarrow 3Na^+(aq.) + PO_4^{3-}(aq)$$

	PO_4^{3-}	+ H_2O	⇌	HPO_4^{2-}	+ OH^-
At equilibrium	C – x	–		x – y	x + y + z

	HPO_4^{2-}	+ H_2O	⇌	$H_2PO_4^-$	+ OH^-
At equilibrium	x – y	–		y – z	x + y + z

	$H_2PO_4^-$	+ H_2O	⇌	H_3PO_4	+ OH^-
At equilibrium	y – z	–		z	x + y + z

$$K_{h_1} = \frac{(x+y+z)(x-y)}{C-x} = \frac{K_w}{K_{a_3}}$$

$$K_{h_2} = \frac{(x+y+z)(y-z)}{x-y} = \frac{K_w}{K_{a_2}}$$

$$K_{h_3} = \frac{(x+y+z)(z)}{y-z} = \frac{K_w}{K_{a_1}}$$

Due to large difference between $K_{h_1} >> K_{h_2} >> K_{h_3}$ neglect of 'z' w.r.t 'y' and 'y' w.r.t 'x'.

$$\therefore \quad K_{h_1} = \frac{x^2}{C-x}; \quad K_{h_2} = y; \quad K_{h_3} = \frac{x.z}{y}$$

$[OH^-] = x + y + z \simeq x$

$[H_3PO_4] = z$

$[HPO_4^{2-}] = x - y \simeq x$

$[H_2PO_4^-] = y - z \cong y$

Case of Cationic Hydrolysis (Polyvalent Cation)

Let us consider a salt solution of $Fe(NO_3)_2$ having concentration 'C' M.
Then its hydrolysis can be studied in similar fashion as above.

$$Fe^{2+} + H_2O \longrightarrow Fe(OH)^+ + H^+$$

$$Fe(OH)^+ + H_2O \longrightarrow Fe(OH)_2 + H^+$$

pH calculation of incomplete salt (Amphiprotic substance) solution

Let us consider a salt solution of NaHA (incomplete salt) having concentration 'C' M.
Its dissociation can be elaborated as:

$$NaHA \longrightarrow Na^+(aq.) + HA^-(aq.)$$
$$-\qquad\qquad C\qquad\qquad C$$

[To form NaHA, H_2A: Weak diprotic acid and NaOH is added]

$$[H_2A + NaOH \rightarrow NaHA + H_2O]$$

Now,

$$HA^-(aq.) + H_2O(l) \rightleftharpoons A^{2-} + H_3O^+$$

Acid

$$K_{a(HA^-)} \text{ or } K_{a_2} = \frac{[A^{2-}][H_3O^+]}{[HA^-]} \qquad ...(1)$$

Base $\quad HA^-(aq.) + H_2O(l) \rightleftharpoons H_2A + OH^-$

$$K_{b(HA^-)} = \frac{K_w}{K_{a_1}} = \frac{[H_2A][OH^-]}{[HA^-]} \qquad ...(2)$$

$$\frac{(2)}{(1)} = \frac{K_w}{K_{a_1}K_{a_2}} = \frac{[H_2A]\,[OH^-]}{[HA^-]} \times \frac{[HA^-]}{[A^{2-}][H_3O^+]}$$

$$\frac{K_w}{K_{a_1}K_{a_2}} = \frac{K_w}{[H_3O^+]^2} \times \frac{[H_2A]}{[A^{2-}]}$$

$$\therefore \quad \boxed{[H_3O^+]^2 = K_{a_1}K_{a_2}\frac{[H_2A]}{[A^{2-}]}}$$

Assumption

Due to very small values of K_{a_1} and K_{a_2} and negligible difference between them.

$[H_2A] = [A^{2-}]$

$\therefore \quad \boxed{[H_3O]^+ = \sqrt{K_{a_1}.K_{a_2}}}$

Case of salt solution of NaH_2PO_4 having concentration 'C' M.

$$NaH_2PO_4 \longrightarrow Na^+ + H_2PO_4^-$$
$$- \qquad\qquad C \quad C$$

$H_2PO_4^-(aq.) + H_2O(l) \rightleftharpoons HPO_4^{2-}(aq.) + H_3O^+(aq.)$
(acid)

$$K_{a_2} = \frac{[HPO_4^{2-}][H_3O^+]}{[H_2PO_4^-]} \qquad ...(3)$$

$H_2PO_4^-(aq.) + H_2O(l) \rightleftharpoons H_3PO_4(aq.) + OH^-(aq.)$

(base)

$$K_b = \frac{K_w}{K_{a_1}} = \frac{[H_3PO_4][OH^-]}{[H_2PO_4^-]} \qquad ...(4)$$

Dividing (4) by (3) and taking assumption: $[HPO_4^{2-}] = [H_3PO_4]$

$$pH = \frac{1}{2}\left(-\log K_{a_1} - \log K_{a_2}\right)$$

Case of salt solution of Na_2HPO_4 having concentration 'C' M.

Behaves as acid: $HPO_4^{2-} + H_2O \rightleftharpoons PO_4^{3-} + H_3O^+$

$$K_{a_3} = \frac{[PO_4^{3-}][H_3O^+]}{[HPO_4^{2-}]}$$

Behaves as base:

$HPO_4^{2-} + H_2O \rightleftharpoons H_2PO_4^- + OH^-(aq.)$

$$K_{b(HPO_4^{2-})} = \frac{K_w}{K_{a_2}} = \frac{[H_2PO_4^-][OH^-]}{[HPO_4^{2-}]}$$

Taking assumption: $[PO_4^{3-}] = [H_2PO_4^-]$

Points to remember

$$\boxed{pH = \frac{1}{2}\left(pK_{a_1} + pK_{a_2}\right)}$$

Points to remember

$$\boxed{pH = \frac{1}{2}\left(pK_{a_1} + pK_{a_2}\right)}$$

Points to remember

$$pH = \frac{1}{2}\left(pK_{a_2} + pK_{a_3}\right)$$

Buffer Solution
Buffer solution formed by two substances.

Definitions

Solutions which resist change in its pH upon addition of small amount of strong acid/base(H^+/OH^-) is called buffer solution.

Types of Buffer solution:
(A) Acidic buffer: Formed by weak acid and salt of weak acid (with strong base) or weak acid and its conjugate.

Examples: Solution made up of CH_3COOH & CH_3COONa, C_6H_5COOH & C_6H_5COONa, HCN & NaCN, HCOOH & HCOOK
Also in conjugate form: C_6H_5COOH & $C_6H_5COO^-$, HCN & CN^- .

(B) Basic buffer: Formed by weak base and salt of weak base (with strong acid) or weak base and it's conjugate.
Examples: Solution made up of NH_4OH or NH_3 & NH_4Cl, RNH_2 & RNH_3Br , $C_6H_5NH_2$ & $C_6H_5NH_3Cl$
Or made up from weak base and its conjugate acid.
Examples: Solution having
NH_3 and NH_4^+
RNH_2 and RNH_3^+
CH_3NH_2 and $CH_3NH_3^+$

2. Buffer solution of single substance:
Salt solution of weak acid and weak base is typical example of such type of buffer.

Example: Solution of CH_3COONH_4, Solution of NH_4CN.

$$pH = \frac{1}{2}(pK_w + pK_a - pK_b)$$

(As discussed earlier.)

pH calculation of Buffer solution
For Acidic Buffer Solution:
Let us consider a solution of CH_3COONa(aq.) having concentration 'C_1' M and CH_3COOH(aq.)having concentration 'C_2' M.
Its analysis can be done as follows:

$$CH_3COONa(aq.) \rightarrow CH_3COO^-(aq.) + Na^+(aq.)$$

$$- \qquad C_1 + x \qquad C_1$$

$$CH_3COOH(aq.) \rightleftharpoons CH_3COO^-(aq.) + H^+(aq.)$$

At equilibrium	$C_2(1-\alpha)$	$C_2\alpha + C_1$	$C_2\alpha$
Or	$C_2 - x$	$x + C_1$	x

$$\therefore \quad K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]} = \frac{(C_1 + x)(x)}{(C_2 - x)}$$

$$\boxed{x = C_2\alpha}$$

If 'x' is negligible w.r.t 'C_2' then $K_a = \frac{(C_1 + x)(x)}{(C_2)}$

If 'x' is negligible w.r.t C_1 then or $\boxed{C_1 \simeq C_2}$ or salt solution fairly concentrated, then

$$\boxed{K_a = \frac{C_1 x}{C_2}}$$

$$[H^+] = x$$

Taking two assumptions:

$$[CH_3COO^-]_{total} = C_1 + x \simeq C_1 \text{ (salt)}$$

$$K_a = \frac{[CH_3COO^-]}{[CH_3COOH]}[H^+]$$

$$[H^+] = K_a \cdot \frac{[CH_3COOH]}{[CH_3COO^-]}$$

$$-\log[H^+] = -\log K_a - \log[CH_3COOH] + \log[CH_3COO^-]$$

(1) $pH = pK_a + \log \frac{[CH_3COO^-]}{[CH_3COOH]}$

(2) $pH = pK_a + \log \frac{[anion]}{[acid]}$

(3) $pH = pK_a + \log \frac{[Conj.\ base]}{[acid]}$

(4) $pH = pK_a + \log \frac{[salt]}{[acid]}$

* Henderson-Hassel balch equations

BUFFER ACTION

Addition of small amount of H^+ from source like HCl/HNO_3:

Let [salt] = 's' and [acid] = 'a' in the buffer solution before addition of H^+.

$$CH_3COO^- + H^+ \longrightarrow CH_3COOH\ (aq.) \qquad (\because K_C = 1/K_a)$$

s	x (L.R)	a
(s – x)	–	a + x

Before addition: $p_1H = pK_a + \log\left(\frac{s}{a}\right)$

After addition: $p_2H = pK_a + \log\left(\frac{s-x}{a+x}\right)$ $(p_2H < p_1H$ because of ratio)

Addition of small amount of OH^- from source like NaOH/KOH:

$$CH_3COOH + OH^- \longrightarrow CH_3COO^- + H_2O\ ; \quad K_C = 1/K_h = 10^{-9}$$

a	y(LR.)	s
a – y	–	s + y

$p_3H = pK_a + \log\left(\frac{s+y}{a-y}\right)$ (slight increase)

Note

Remember: [s] + [a] is always constant.

pH Calculation of Basic Buffer Solution:

Let us consider a solution of NH_4Cl(aq.) and NH_4OH (aq.) or NH_3 having concentration C_1 and C_2 respectively.

$$NH_4Cl(aq.) \longrightarrow NH_4^+(aq.) + Cl^-(aq.)$$

$$C_1 + x \qquad C_1$$

$$NH_3(aq.) + H_2O(l) \rightleftharpoons NH_4^+(aq.) + OH^-(aq.)$$

$$C_2(1-\alpha) \qquad C_2\alpha + C_1 \qquad C_2\alpha$$

$$\text{Or, } C_2 - x \qquad C_1 + x \qquad x$$

$$\therefore \quad K_b = \frac{(C_1 + x)(x)}{(C_2 - x)}$$

If 'x' is negligible w.r.t 'C_1' then $K_b = \dfrac{(C_1 + x)(x)}{C_2}$

If $C_1 = C_2$, or salt solution is fairly concentrated, then $\boxed{K_b = \dfrac{C_1}{C_2}x}$

As, $[OH^-] = x$

$[NH_4^+] \simeq [NH_4Cl] \simeq [\text{salt}]$

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_3]}; \quad [OH^-] = \frac{[NH_3]K_b}{[NH_4^+]}$$

$$-\log[OH^-] = -\log K_b - \log[NH_3] + \log[NH_4^+]$$

$$pOH = pK_b + \log\frac{[NH_4^+]}{[NH_3]} = pK_a + \log\frac{[\text{Conj. acid}]}{[\text{Base}]}$$

$$\therefore \; 14 - pH = pK_b + \log\frac{[\text{cation}]}{[\text{base}]} = pK_b + \log\frac{[\text{salt}]}{[\text{base}]}.$$

The above formula applied when there is same concentration of salt and cation.

And not in this type like $\boxed{\begin{matrix}(NH_4)_2SO_4 : C \\ NH_4^+ : 2C\end{matrix}}$

Buffer Action

As from above : $pOH = pK_b + \log\left(\dfrac{s}{b}\right)$ where [salt] = s, [base] = b in buffer solution earlier.

(A) H_3O^+/H^+ addition (in small quantity):

$NH_3(aq.) + H^+(aq.) \longrightarrow NH_4^+(aq.)$ $\boxed{K_c = 1/K_a}$

b	x(L.R.)	s
b-x	–	s + x

After addition, $\uparrow \boxed{p_2OH} = pK_b + \log \frac{(s+x)}{b-x}$

(B) OH^- addition (in small quantity):

$NH_4^+(aq.) + OH^-(aq.) \longrightarrow NH_3(aq.) + H_2O(l)$ $\boxed{K_c = 1/K_b}$

s	y(L.R.)	b
(s-y)	–	(b+y)

$\boxed{\because pH \uparrow}$ $\downarrow p_2OH = pK_a + \log \frac{(s-y)}{b+y}$

Addition of $H_2O(l)$ in Buffer Solution (Effect of Dilution)

Addition of water in small quantity shows no change in pH (appreciably) of buffer solution.

Addition of water in large quantity results in pH change of the solution and tends to 7.

i.e $pH \rightarrow 7$

$CH_3COOH\ (aq.) \rightleftharpoons CH_3COO^-(aq.) + H^+\ (aq.)$

At equilibrium $(C_2 - x)$ $(C_1 + x)$ (x)

And after dilution,

At equilibrium $(C_2' - x')$ $(C_1' + x')$ (x')

$$(\text{Constant})K_a = \frac{(C_1' + x')x'}{(C_2' - x')}$$

Useful BUFFER RANGE (for industrial application)

For acidic buffer

$$\frac{[\text{salt}]}{[\text{acid}]} = \frac{1}{10} \text{ to } \frac{10}{1}$$

$\boxed{pH = pK_a \pm 1}$

For basic buffer

$$\frac{[\text{salt}]}{[\text{base}]} = \frac{1}{10} \text{ to } \frac{10}{1}$$

$\boxed{pOH = pK_b \pm 1}$

Buffer solution is at its maximum capacity when ;

[Salt] = [Acid] [Salt] = [Base]

$\boxed{pH = pK_a}$ $\boxed{pOH = pK_b}$

Proof: For Acidic Buffer Solution

$$p_1H = pK_a + \log\left(\frac{s}{a}\right)$$

Δb: small amount of strong base is added to buffer solution.

$$p_2H = pK_a + \log\left(\frac{s+\Delta b}{a-\Delta b}\right)$$

Change in pH = $\Delta pH = p_2H - p_1H = \log\left(\frac{(s+\Delta b)}{(a-\Delta b)} \times \frac{a}{s}\right)$

$$\Delta pH = \frac{1}{2.303}\ln\left(\frac{1+(\Delta b/s)}{1-(\Delta b/a)}\right) = \frac{1}{2.303}\left(\frac{\Delta b}{s} - \left(-\frac{\Delta b}{a}\right)\right)$$

$\Delta pH = \frac{\Delta b}{2.303}\left(\frac{a+s}{as}\right)$; Let a + S = x; $\Delta pH = \frac{\Delta b}{2.303}\left[\frac{x}{a(x-a)}\right]$

Buffer capacity in terms of buffer index = $\frac{\Delta b}{\Delta pH} = 2.303\left(\frac{a.s}{a+s}\right)$; ln 10 = 2.303

$\boxed{\ln e = 2.303(\ln 10)}$

$\boxed{\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - ....}$

Buffer capacity = $\boxed{\frac{\text{Moles of } H^+ / OH^- \text{ added per litre of buffer solution}}{\text{Chang in pH}}}$

For maximum buffer capacity :

$$\frac{d}{da}(\Delta b / \Delta pH) = 0$$

$$x - 2a = 0$$

$$\boxed{\frac{x}{2} = a}$$

$\therefore$ $\boxed{s = x - a = \frac{x}{2}}$

$\therefore$ At maximum buffer capacity: $\boxed{[Salt] = [Acid]}$

Indicators

General acidic indicator: HIn

General basic indicator: InOH

$InOH(aq.) \rightleftharpoons In^{+}(aq.) + OH^{-}(aq.)$

Also, $\underset{\substack{\text{Unionised form}\\ \text{or acidic form}\\ \text{(Colour I)}}}{HIn} \underset{H^{+}}{\overset{OH^{-}}{\rightleftharpoons}} H^{+} + \underset{\substack{\text{Ionised form or}\\ \text{basic form}\\ \text{(Colour II)}}}{In^{-}}$

$$K_{HIn} \text{ or } K_{IN} = \frac{[H^{+}][In^{-}]}{[HIn]};$$

$$[H^{+}] = \frac{K_{In} \cdot [HIn]}{[In^{-}]}$$

$$pH = pK_{In} + \log\left[\frac{[In^{-}]}{[HIn]}\right]$$

$$\% \text{ ionised form of indicator} = \frac{[In^{-}]}{[HIn] + [In^{-}]} \times 100$$

$$\% \text{ unionised form of indicator} = \frac{[HIn]}{[HIn] + [In^{-}]} \times 100$$

Colour Transition Range (C.T.R)

In general: $\frac{[In^{-}]}{[HIn]} = \frac{1}{10} \text{ to } \frac{10}{1}$

$$\boxed{pH = pK_{In} \pm 1}$$

If pH of solution $\geq (pK_{In} + 1)$ then solution has characteristics of In^{-}.

If pH of solution $\leq (pK_{In} - 1)$ then solution has characteristics of HIn.

If pH of solution exists between $(pK_{In} - 1)$ to $(pK_{In} + 1)$ then indicator changes its colour.

Phenolphthalein (HpH):

$\underset{\substack{\text{ionised or}\\ \text{acidic form}\\ \text{(Colourless)}}}{HpH} \underset{H^{+}}{\overset{OH^{-}}{\rightleftharpoons}} H^{+} + \underset{\substack{\text{Ionised or}\\ \text{basic form}\\ \text{(pink)}}}{pH^{-}}$

Definitions

In general indicators are either weak organic acids or weak organic bases having different colour characteristics in ionised and unionised form

Note

Here, Total H^{+} of solution is taken in which indicator is added.

Note

Indicators change its colour in certain pH range. This pH range is called colour transition range.

Points to remember

It is an acidic indicator.

C.T.R.: 8.3 to 10

Indicaotr	Colour I	Colour II
Phenolphthalein	Colourless	Pink
Methyl orange	Red	Yellow

If pH of solution ≥ 10 then solution has basic form colour i.e pink.
If pH of solution ≤ 8.3 then solution has acidic form colour i.e colourless.
If pH of solution is between 8.3 and 10, then indicator changes its colour.

Methyl Orange (MeOH):

$$\underset{\substack{\text{Unionised form or}\\ \text{basic form}\\ \text{(yellow)}}}{MeOH(aq.)} \underset{OH^-}{\overset{H^+}{\rightleftharpoons}} Me^+ + \underset{\substack{\text{Ionised form or}\\ \text{acidic form}\\ \text{(red)}}}{OH^-}$$

If pH of solution ≥ 4.4 then solution has basic form colour i.e yellow colour.
If pH of solution ≤ 3.1 then solution has acidic form colour i.e red colour.
If pH of solution is between 3.1 to 4.4, then indicator changes its colour.

Points to remember

It is a basic indicator.
C.T.R. : 3.1 to 4.4

Note

Better indicator is one whose C.T.R. is less!

MASS BALANCE: Based on POAC
This method is generally used by scientists.

1. Suppose we have a weak acid of type HA(aq.) having concentration 'C' M.

$$HA\ (aq.) \rightleftharpoons H^+(aq.) + A^-(aq.)$$

t = 0	C	–	–
At equilibrium	C - x	x	x

POAC on 'A':

$$[HA]_{t=0} \times 1 = [HA]_{t=eqm} \times 1 + [A^-]_{t=eqm} \times 1$$

C = (C – x) + x
Hence Proved.

2. Let us consider another weak diprotic acidic solution H_2A having concentration 'C' M.

$$H_2A(aq.) \rightleftharpoons H^+(aq.) + HA^-(aq.)$$

C-x x+y x–y

$$HA^-(aq.) \rightleftharpoons H^+(aq.) + A^{2-}(aq.)$$

x–y x+y y

POAC on 'A':

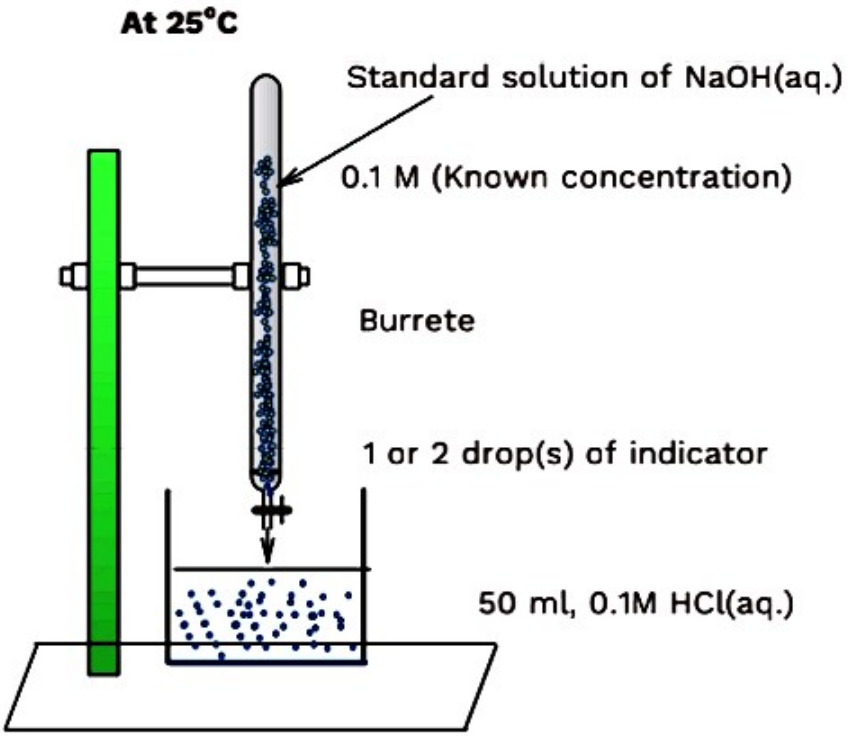
$$[H_2A]_{t=0} \times 1 = [H_2A]_{t=eqm} \times 1 + [HA^-]_{t=eqm} \times 1 + [A^{2-}]_{t=eqm} \times 1$$

$C = (C - x) + (x - y) + y$

Hence Proved.

Titration

Titration of strong acid with strong base



V_{NaOH} = 0 ml; [H+] = 0.1 M pH = −log (0.1⁻) = −log (10^{-1}) = 1	$HCl(aq.) + NaOH(aq.) \longrightarrow NaCl(aq.) + H_2O(l)$ Millimoles 5 1 4 – 1 $[H^+]_{Resultant} = \frac{\text{Millimoles of HCl} - \text{Millimoles of NaOH}}{\text{Total volume of solution (in ml)}}$ $[H^+] = \frac{5-1}{50+10} = \frac{1}{15}M$ pH ≃ 1.17
V_{NaOH} = 10 ml; 0.1M V_{NaOH} = 40 ml; 0.1M V_{NaOH} = 49 ml, 0.1M V_{NaOH} = 49.9 ml; 0.1M V_{NaOH} = 50 ml; 0.1M	$[H^+] = \frac{5-4}{50+40} = \frac{1}{90}M$ pH ≃ 1.95 $[H^+]_{Resultant} = \frac{5-4.9}{50+49} = \frac{0.1}{99} = 10^{-3}M$ pH = 3 $[H^+]_{Resultant} = \frac{5-4.99}{50+49.9} = 10^{-4}M$ pH = 4 At neutralisation point (at equivalent point) Millimoles of HCl = Millimoles of NaOH pH = 7

HCl is completely neutralised by base, NaCl is formed.

$$[NaCl] = \frac{\text{Millimoles of acid}}{\text{Total volume}} = \frac{5}{100} = 0.05\ M$$

$HCl(aq.) + NaOH(aq.) \longrightarrow NaCl(aq.) + H_2O(l)$

Millimoles 5 5.01

– 0.01 5

V_{NaOH} = 51 ml;	$[OH^-]_{Resultant} = \frac{\text{Millimoles of NaOH} - \text{Millimoles of HCl}}{\text{Total volume of solution (in ml)}}$ $= \frac{5.01 - 5}{50 + 50.1} = 10^{-4}M$; pOH = 4 $\boxed{pH = 10}$

C.T.R

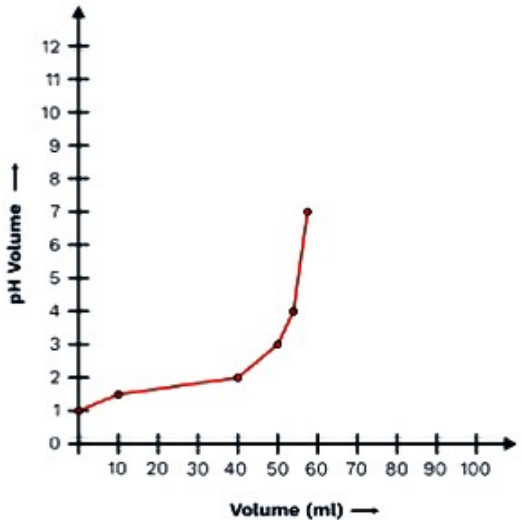
Methyl Orange (MeOH): = 3.1 to 4.4

Bromothymol blue : 6 to 7.6

HPh : 8.3 to 10

All three indicators are suitable for this titration (S.A. with S.B.)

Bromothymol blue is best.



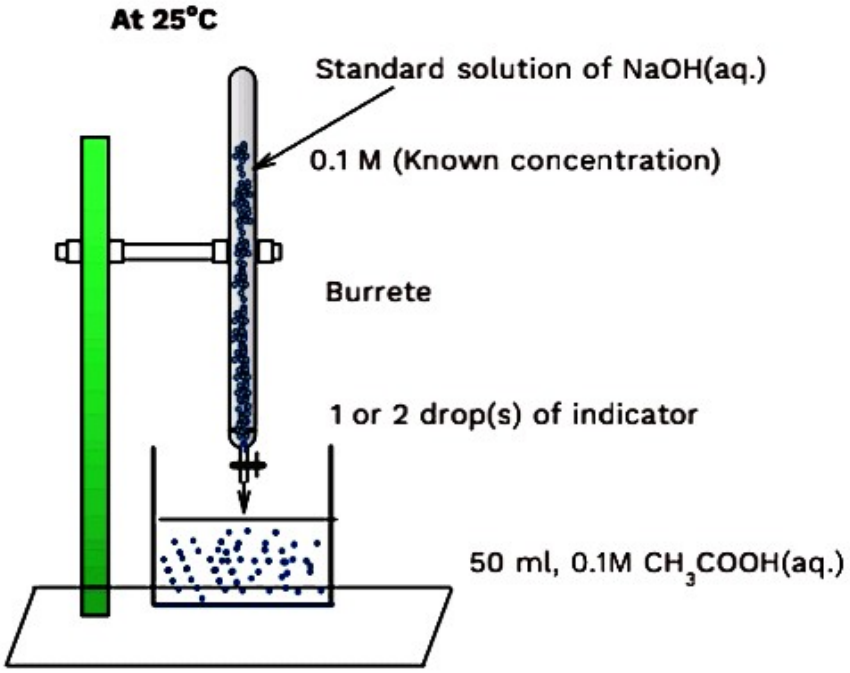
Concept Ladder

Selection of Indicator

C.T.R of an indicator should exist in **flat region** of titration curve.

pH of solution at neutralisation point = mid point of C.T.R. of indicator.

Weak Acid V/s Strong Base



0.1 M, 50 ml

CH_3COOH(aq.) solution

$K_a = 1.8\times10^{-5}$

Case I

- $V_{NaOH} = 0$ ml

$$pH = \frac{1}{2}(pK_a - \log C)$$

$$pH = \frac{1}{2}\left(4.74 - \log 10^{-1}\right)$$

$$pH = \frac{1}{2}(5.74) = 2.87$$

Case II

- $V_{NaOH} = 10$ ml

$$pH = pK_a + \log\frac{s}{a}$$

$$pH = 4.74 + \log\frac{1}{4}$$

$$pH = 4.137$$

	$CH_3COOH(aq.)$	+ $NaOH(aq.)$	⟶ $CH_3COONa(aq.)$	+ $H_2O(l)$
millimoles	5	1	-	-
millimoles	4	-	1	-

Case III

- V_{NaOH} = 20 ml

$$pH = pK_a + \log\frac{s}{a}$$

$$pH = 4.74 + \log\frac{2}{3}$$

$$pH = 4.576$$

Note
Buffer solution is at its maximum capacity.

Case IV

- V_{NaOH} = 25 ml

$$pH = pK_a = 4.74$$

millimoles of $CH_3COOH(aq.)$ after reaction = millimoles of NaOH(aq.) after reaction

Case V

- V_{NaOH} = 30 ml

$$pH = pK_a + \log\frac{s}{a}$$

$$pH = 4.74 + \log\frac{3}{2}$$

$$pH = 4.916$$

Case VI

- V_{NaOH} = 40 ml

$$pH = pK_a + \log\frac{s}{a}$$

$$pH = 4.74 + \log\frac{4}{1}$$

$$pH = 5.34$$

Case VII

- V_{NaOH} = 49 ml

$$pH = pK_a + \log\frac{s}{a}$$

$$pH = 4.74 + \log\frac{4.9}{0.1}$$

$$pH = 6.43$$

Case VIII

- V_{NaOH} = 50 ml, $V_{resultant\ solution}$ = 100 ml

  Anionic hydrolysis takes place.

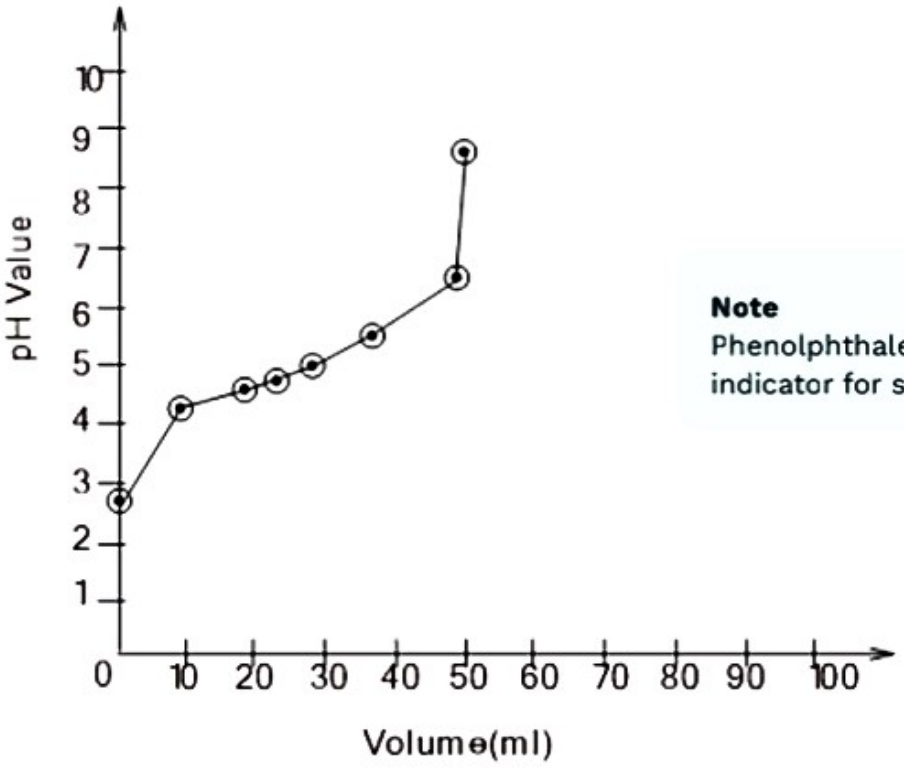
  $$14 - pH = \frac{1}{2}[pK_w - pK_a - \log C]$$

  $$14 - pH = \frac{1}{2}[14 - 4.74 - \log 5\times10^{-2}]$$

  $$pH = 14 - 5.28$$

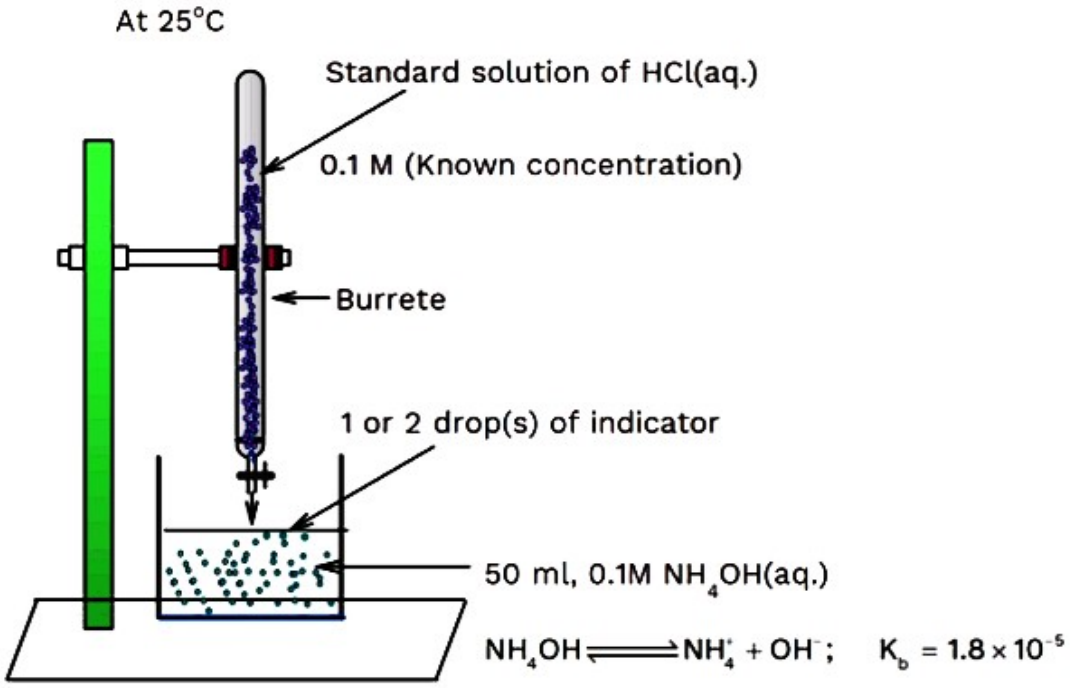
  $$pH = 8.7$$

Volume(ml)	0	10	20	25	30	40	49	50
pH	2.87	4.137	4.576	4.74	4.916	5.34	6.43	8.7



Note
Phenolphthalein is suitable indicator for such titration.

Strong Acid V/s Weak Base



Case I * $V_{HCl} = 0$ ml

$$K_b = \frac{[OH^-][NH_4^+]}{[NH_4OH]}$$

$$K_b = \frac{C\alpha^2}{1-\alpha} \text{ if } \sqrt{\frac{K_a}{C}} < 0.05$$

$$K_b = C\alpha^2$$

$$\alpha = \sqrt{\frac{K_b}{C}}$$

$$[OH^-] = C\alpha$$

$$[OH^-] = C\sqrt{\frac{K_b}{C}}$$

$$[OH^-] = \sqrt{K_b C}$$

$$pOH = \frac{1}{2}(pK_b - \log C)$$

Note
At 25°C
pH ⇒ 11.13

$$pOH = \frac{1}{2}\left(4.74 - \log 10^{-1}\right)$$

$$\Rightarrow pOH = \frac{4.74 + 1}{2}$$

$$\Rightarrow pOH = \frac{5.74}{2} = 2.87$$

$$pH = 14 - 2.87 = 11.13$$

Case II

* $V_{HCl} = 10$ ml , 0.1 M

$V_{soln} = 60$ ml

$$pOH = pK_b + \log \frac{s}{b}$$

$$pOH = 4.74 + \log \frac{1}{4}$$

$$pOH = 4.1$$

$$pH = 9.9$$

	$NH_4OH(aq.)$		$NH_4Cl(aq.)$	
	$NH_4OH + HCl\,(aq) \longrightarrow$		$NH_4Cl + H_2O\,(l)$	
millimoles	5	1	-	-

Case III

* $V_{HCl} = 20$ ml, 0.1 M

$V_{solution.} = 70$ ml

$$pHO = pK_b + \log \frac{s}{b}$$

$$pHO = 4.74 + \log \frac{2}{3}$$

$$pHO = 4.56$$

$$pH = 9.44$$

Note
Basic buffer solution is formed.

Case IV

* V_{HCl} = 25 ml, 0.1 M

$V_{solution}$ = 50 ml

$pOH = pK_b = 4.74$

$pn = 14 - 4.74 = 9.26$

Case V

* V_{HCl} = 30 ml, 0.1 M

$V_{solution}$ = 80 ml

$$pOH = pK_b + \log \frac{s}{b}$$

$$pOH = 4.74 + \log \frac{2}{3}$$

$pOH = 4.56$

$pH = 9.44$

Case VI

* V_{HCl} = 40 ml, 0.1 M

$V_{solution}$ = 90 ml

$$pOH = 4.74 + \log \frac{4}{1}$$

$pOH = 5.34$

$pH = 8.66$

Case VII

* V_{HCl} = 49 ml, 0.1 M

$V_{solution}$ = 99 ml

$$pOH = 4.74 + \log \frac{4.9}{0.1}$$

$pOH = 6.43$

$pH = 7.47$

Note
Cationic hydrolysis takes place.

millimoles of HCl after reaction = millimoles of NH_4OH after reaction

Note
Methyl orange is suitable indicator for this type of titration.

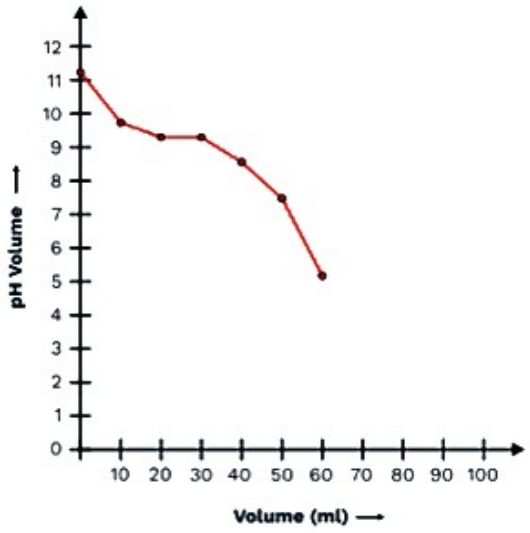
Case VIII * V_{HCl} = 50 ml, $V_{resultant\ solution}$ = 100 ml

V_{sol^n} = 100 ml

$$pH = \frac{1}{2}(pK_w - pK_b - \log C)$$

pH = 5.28

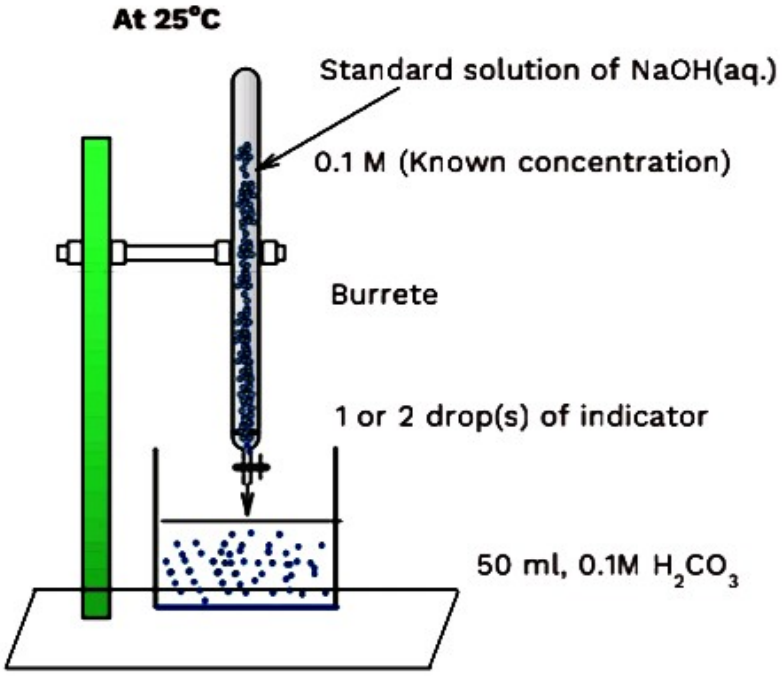
Volume(ml)	0	10	20	30	40	49	50
pH	11.13	9.9	9.44	9.44	8.66	7.47	5.28



Weak acid V/s Weak base

* No sharp change at equivalent point.

Titration of H_2CO_3 V/s NaOH



Case I * V_{NaOH} = 0 ml

$$\begin{matrix} H_2CO_3\,(aq.) & \rightleftharpoons & H^+\,(aq.) & + & HCO_3^-\,(aq.) \\ C-x & & x+y & & x-y \end{matrix}\Bigg]; \quad K_{a_1} = \frac{(x+y)(x-y)}{C-x}$$

$$\begin{matrix} HCO_3^- & \rightleftharpoons & H^+ & + & CO_3^{2-} \\ x-y & & x+y & & y \end{matrix}\Bigg]; \quad K_{a_2} = \frac{(x+y)}{x-y}y$$

$$10^{-5} = \frac{x.x}{0.1-x}$$

$$10^{-5} = \frac{x^2}{0.1}$$

$$x^2 = 10^{-6}$$

$$x = 10^{-3}$$

$$10^{-9} = \frac{x}{x}y$$

$$10^{-9} = y$$

$$K_{a_2} = y$$

pH = 3

Case II * V_{NaOH} = 10 ml

m. moles = 1

Volume of mixture solution = 60 ml

$$pH = pK_{a_1} + \log\frac{1}{4}$$

$$pH = 5 + \log 0.25$$

pH = 4.39

Case III • $V_{NaOH} = 20\,ml$ m.moles = 2

Volume of mixture solution = 70 ml

$$pH = pK_{a_1} + \log\frac{2}{3}$$

$$pH = 5 + \log 0.66$$

$$pH = 4.82$$

Case IV • $V_{NaOH} = 30\,ml$ m.moles = 3

Volume of mixture solution = 80 ml

$$pH = pKa_1 + \log\frac{3}{2}$$

$$pH = 5 + \log 1.5$$

$$pH = 5.17$$

Case V • $V_{NaOH} = 40\,ml$ Volume of mixture solution = 80 ml

$$pH = pK_{a_1} + \log\frac{4}{1}$$

$$pH = 5 + \log 4$$

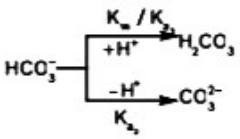
$$pH = 5.6$$

Case VI • $V_{NaOH} = 50\,ml$ m .moles = 5

Volume of mixture solution = 100 ml

pH = 6.8 [I^st neutraisation]

	H_2CO_3	+ NaOH ⟶	HCO_3^-	+ H_2O
m.moles	5	5	–	–
m.moles	–	–	5	–



$$pH = \frac{1}{2}\left[pK_{a_1} + pK_{a_2}\right]$$

2nd Neutralisation:

Case VI . 1 * $V_{NaOH} = 60\,ml$

$$pH = pK_{a_2} + \log 0.25$$

$$pH = 9 + \log 0.25 = 8.39$$

Case VI . 2 * $V_{NaOH} = 70\,ml$

$$pH = pK_{a_2} + \log \frac{2}{3}$$

$$pH = 9 + \log 0.66 = 8.82$$

Case VI . 3 * $V_{NaOH} = 80\,ml$

$$pH = 9 + \log 1.5$$

$$pH = 9.17$$

Case VI . 4 * $V_{NaOH} = 90\,ml$

$$pH = 9 + \log \frac{4}{1}$$

$$pH = 9.6$$

Case VI . 5 * $V_{NaOH} = 100\,ml$

$$pH = 9 + \log \frac{4}{1}$$

$$pH = 9.6$$

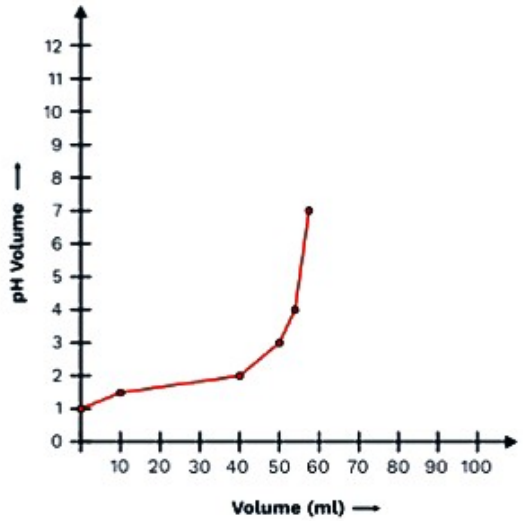
	$NaHCO_3 + NaOH \longrightarrow$		$Na_2CO_3 + H_2O$	
millimoles	5	1	-	-
millimoles	4	-	1	-

$$[Na_2CO_3] = \frac{5}{100} = \frac{1}{20} M$$

Do You Know?

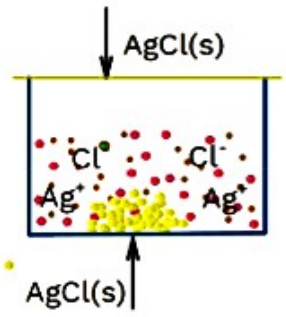
From here hydrolysis of carbonate ion will takes place and pH of solution after 2nd equivalent point is calculated on the basis of hydrolysis of carbonate ion.

Volume(ml)	0	10	20	30	40	50	60	70	80	90	100
pH	3	4.39	4.82	5.17	5.6	6.8	8.39	8.82	9.17	9.6	9.6



Solubility

AgCl is sparingly soluble substance (S.S.S.) which establishes heterogeneous equilibrium.



$$AgCl(s) \underset{R.D.P}{\overset{R.O.D.}{\rightleftharpoons}} Ag^{+}(aq.) + Cl^{-}(aq.)$$

At eqm. _ s s

Rate of dissolution of AgCl = Rate of precipitation of ions.

Definitions

- Soluble substances are those which have solubility concentration > 0.1 M.
- Partial or slightly soluble substances are those which have solubility concentration = 0.01 to 0.1 M.
- Sparingly soluble substances are those which have solubility concentration < 0.01 M.
- (Solubility is very less in water)

Let solubility of AgCl be 's'(mol/L) At eqm.

$AgCl(aq.)$: s :- $AgCl(aq.) \longrightarrow Ag^+(aq.) + Cl^-(aq.)$

	–	s	s

$AgCl(s) \longrightarrow Ag^+(aq.) + Cl^-(aq.)$

	s	s

K_c or $K_{SP} = [Ag^+(aq.)]\,[Cl^-(aq.)]$

Note

K_{SP} (Solubility Product) = s.s;

$K_{SP} = s^2$

$\boxed{s = \sqrt{K_{SP}}}$ $\boxed{s : mol/L}$

Ionic product or Reaction Quotient (Q_c or Q)

$Q = [Ag^+(aq.)]\,[Cl^-(aq.)]$ (Solution at time 't' not necessary at equilibrium)

Points to remember

- If $Q < K_{SP}$, then solution is unsaturated (No precipitation takes place)
- If $Q = K_{SP}$ then solution is saturated (Reaction is at equilibrium)
- If $Q > K_{SP}$, then solution in oversaturated or supersaturated (Precipitation takes place

Q11 **If 10^{-5} M $AgNO_3$(aq.) solution is mixed with 10^{-4} M KCl(aq.) solution in volume ratio of 4:1. Find whether precipitation takes place or not (Given: K_{SP}(AgCl) = 10^{-10})**

Sol. Let volume of KCl(aq.) solution is V ml; then volume of $AgNO_3$ = 4V ml

Total volume of mixture solution = 5V

$$[Ag^+]_{Resultant} = \frac{10^{-5} \times 4V}{4V + V} = 0.8 \times 10^{-5}$$

$$[Cl^-]_{Resultant} = \frac{10^{-4} \times V}{5V} = 0.2 \times 10^{-4}$$

$Q = [Ag^+][Cl^-] = 0.16 \times 10^{-9} = 1.6 \times 10^{-10} > K_{SP}$

Precipitation takes place.

Case of percentage ionisation of sparingly soluble substance:

Let us assume a sparingly soluble salt of type AB(s).

Assumption: Dissolved AB is 95% ionised.

$$AB(s) \rightarrow AB(aq.) \rightarrow A^+(aq.) + B^-(aq.)$$

$$AB(s) \rightleftharpoons A^+(aq.) + B^-(aq.)$$

$$AB(s) \rightleftharpoons A^+(aq.) + B^-(aq.)$$

At eqm. – 0.95s 0.95s

$K_{SP} = (0.95s)^2$

where Solubility of AB(s) : s

$$AB(aq.) \rightleftharpoons A^+(aq.) + B^-(aq.)$$

At eqm.: 0.05s 0.95s 0.95s

Relation between solubility (s) and solubility product (K_{sp}):

For 'AB' Type S.S.S:	Example AgBr, AgI, $BaSO_4$ etc. Let solubility of $BaSO_4$ is 's'. $BaSO_4(s) \rightleftharpoons Ba^{2+}(aq.) + SO_4^{2-}(aq.)$ At eqm. – s s **Note** $K_{SP} = [Ba^{2+}(aq.)]\,[SO_4^{2-}(aq.)]$ $\Rightarrow K_{SP} = s^2$
For A_2B/ AB_2 Type S.S.S:	Examples: Ag_2CrO_4, PbI_2, $Fe(OH)_2$ etc. Let solubility of Ag_2CrO_4 is 's'. $Ag_2CrO_4(s) \rightleftharpoons 2Ag^+(aq.) + CrO_4^{2-}(aq.)$ At eqm. – 2s s **Note** $K_{SP} = [Ag^+(aq.)]^2[CrO_4^{2-}(aq.)]$ $= (2s)^2.s = 4s^3$

For A_3B/AB_3 Type S.S.S:	Examples: Ag_3PO_4, $Fe(OH)_3$ etc. Let solubility of $Fe(OH)_3$ is 's'. $Fe(OH)_3(s) \rightleftharpoons Fe^{3+}(aq.) + 3OH^-(aq.)$ At eqm. – s 3s **Note** $K_{SP} = [Fe^{3+}][OH^-]^3$ $= s(3s)^3 = 27s^4$
For A_xB_y Type S.S.S:	Let solubility of A_xB_y is 's'. $A_xB_y(s) \rightleftharpoons xA^{y+}(aq.) + yB^{x-}(aq.)$ At eqm. – xs ys $K_{SP} = (xs)^x(ys)^y$ **Note** $\boxed{K_{SP} = x^x.y^y.s^{(x+y)}}$

Q12 **The relation between solubility and solubility product for $Sr_3(PO_4)_2$ is:**

Sol. $K_{SP} = (3s)^3(2s)^2 = \boxed{K_{SP} = 108\,s^5}$

Solubility of AgCl in water:

$AgCl(s) \rightleftharpoons Ag^+(aq.) + Cl^-(aq.)$

At eqm. – s s

$\boxed{K_{SP} = s^2}$ where 's' is in $mol.L^{-1}$

Given: $K_{SP}(AgCl) = 10^{-10}$

As, $K_{SP} = s^2$

Therefore , $s = \sqrt{K_{SP}}$

$\Rightarrow s = \sqrt{10^{-10}}$

$\Rightarrow s = 10^{-5}$

Solubility of AgCl in aqueous solution of $AgNO_3$:

Let $AgNO_3$(aq.) have concentration 'C' M in which sparingly soluble salt AgCl is being dissolved.

Let solubility of AgCl in aqueous solution of $AgNO_3$ in 'x'.

$$AgCl(s) \rightleftharpoons Ag^+(aq.) + Cl^-(aq.)$$

At eqm. – $(x + C)$ x

This equilibrium is established in $AgNO_3$ (aq.) solution.

$$K_{SP} = [Ag^+(aq.)]_{total}[Cl^-(aq.)]$$

$$K_{SP} = (x + C)x$$

where $x < s$

Due to common ion effect

('s' is solubility of AgCl in water)

Q13 **Calculate solubility of AgCl in 0.01 M $AgNO_3$(aq.) solution. [Given K_{SP}(AgCl) = 10^{-10}]**

Sol. $K_{SP} = (x + 0.01)x$ where $x < 10^{-5}$

$10^{-10} = 0.01\ x$ (neglect 'x' w.r.t 0.01)

$x = 10^{-8}$ M

whereas 's' = 10^{-5} M in water.

Simultaneous Solubility

Examples of simultaneous solubility are mixture of following solutions:

AgCl and AgBr

$BaSO_4$ and $CaSO_4$

Ag_2S and Ag_2CrO_4

$Sr(OH)_2$ and $Fe(OH)_2$ etc.

$BaSO_4(s) \rightleftharpoons Ba^{2+}(aq.) + SO_4^{2-}(aq.)$ $\boxed{BaSO_4\ K_{SP_1} = s_1^2\ \text{In pure water}}$

$CaSO_4(s) \rightleftharpoons Ca^{2+}(aq.) + SO_4^{2-}(aq.)$ $\boxed{CaSO_4\ K_{SP_2} = s_2^2\ \text{In pure water}}$

In absence of each other

Let solubility of $BaSO_4$ is 'x' and $CaSO_4$ is 'y' in presence of each other.

where $x < s_1$ and $y < s_2$ due to common ion effect of SO_4^{2-} ion. So, both the equilibrium

will move backward in
presence of each other.

$$BaSO_4(s) \rightleftharpoons Ba^{2+}(aq.) + SO_4^{2-}(aq.),\ K_{SP_1}$$

At eqm. – x (x + y)

$$CaSO_4(s) \rightleftharpoons Ca^{2+}(aq.) + SO_4^{2-}(aq.),\ K_{SP_2}$$

At eqm. – y (y + x)

$K_{SP_1} = [Ba^{2+}(aq.)][SO_4^{2-}]_{total}$; $K_{SP_2} = [Ca^{2+}(aq.)][SO_4^{2-}]_{total}$

$K_{SP_1} = x(x + y)$ ...(i) ; $K_{SP_2} = y(x + y)$ ...(ii)

On adding equation (i) and equation (ii):

$$(x + y)^2 = K_{SP_1} + K_{SP_2}$$

$$\left[SO_4^{2-}\right]_{total} \Rightarrow x + y = \sqrt{K_{SP_1} + K_{SP_2}}$$

$$\frac{\left[Ba^{2+}\right]}{\left[Ca^{2+}\right]} = \frac{x}{y} = \frac{K_{SP_1}}{K_{SP_2}}$$

Q14 **Let the solubilities of AgCl in pure water, 0.01 M $CaCl_2$, 0.01 M NaCl and 0.04 M $AgNO_3$ be s_1, s_2, s_3 and s_4 respectively. What is the correct order?**

(A) $s_1 > s_2 > s_3 > s_4$ **(B)** $s_1 > s_2 = s_3 > s_4$

(C) $s_1 > s_3 > s_2 > s_4$ **(D)** $s_4 < s_2 > s_3 > s_1$

Sol. C

From $CaCl_2$: $[Cl^-] = 0.01 \times 2 = 0.02$ M

From NaCl: $[Cl^-] = 0.01$ M

From $AgNO_3$: $[Ag^+] = 0.04$ M

$s_1 > s_3 > s_2 > s_4$ (Due to common ion effect)

Q15 **50 litre of a solution containing 10^{-5} mole of Ag^+ is mixed with 50 litre of a 2×10^{-7} M HBr solution. $[Ag^+]$ in resultant solution is: [Given: K_{sp} (AgBr) = 5×10^{-13}]**

(A) 10^{-5} M **(B)** 10^{-6} M **(C)** 10^{-7} M **(D)** None of these

Sol. $V_{total} = 50 + 50 = 100$ L;
Now after mixing:

$$[HBr] = \frac{M \times V}{2V} = \frac{2 \times 10^{-7} \times 50}{100}$$

$[HBr]_{resultant} = 10^{-7}$ M
$[Br^-]_{resultant} = 10^{-7}$ M

$$[Ag^+]_{resultant} = \frac{10^{-5}}{100} = 10^{-7}M$$

$Q = [Ag^+]_{resultant}\,[Br^-]_{resultant} = 10^{-7} \times 10^{-7} = 10^{-14}$
As: $Q < K_{SP}$;
Hence, no precipitation takes place.

Q16 **The solubilities of AgCl and AgBr in water are 10^{-5} and 10^{-6} mol/L. If an equi-molar mixture of AgCl and AgBr, are added into water then what is the ratio of $[Br^-]/[Cl^-]$ in resultant solution.**
(A) 100 : 1 **(B) 1 : 100** **(C) 1 : 99** **(D) 99 : 1**

Sol. B

$AgCl(s) \rightleftharpoons Ag^+(aq.) + Cl^-(aq.)$; $K_{SP_1} = \left(10^{-5}\right)^2 = 10^{-10}$

$AgBr(s) \rightleftharpoons Ag^+(aq) + Br^-(aq)$; $K_{SP_2} = \left(10^{-6}\right)^2 = 10^{-12}$

$$\frac{K_{SP_2}}{K_{SP_1}} = \frac{[Br^-]}{[Cl^-]} = \frac{10^{-12}}{10^{-10}} = \frac{10^{-2}}{1} = \frac{1}{100}$$

Q17 **pH of a saturted solution of silver salt of monobasic acid HA is found to be 9. Find the K_{SP} of sparingly soluble salt AgA(s). Given: $K_a(HA) = 10^{-10}$.**
(A) 1.1×10^{-11} **(B) 1.1×10^{-10}** **(C) 10^{-12}** **(D) None of these**

Sol. A

$AgA(s) \rightleftharpoons Ag^+(aq.) + A^-(aq.)$ $K_{SP} = x(x-y)$

At eqm. – x (x – y)

$A^-(aq.) + H_2O(l) \rightleftharpoons HA(aq.) + OH^-(aq.)$

At eqm. x – y – y y

$$K_h = \frac{K_w}{K_a} = \frac{y^2}{(x-y)}$$

$$\frac{10^{-14}}{10^{-10}} \Rightarrow 10^{-4} = \frac{(10^{-5})^2}{x - 10^{-5}}$$

$x - 10^{-5} = 10^{-6}$

$x = 10^{-5} + 10^{-6} = 1.1 \times 10^{-5}$ M

$K_{SP} = 1.1 \times 10^{-5} \times 0.1 \times 10^{-5} = 1.1 \times 10^{-11}$

Solubility of AgCN in water:

Let solubility of AgCN in pure water is 'x'.

$$AgCN(s) \rightleftharpoons Ag^+(aq.) + CN^-(aq.);\ K_{SP}$$

At eqm. – x (x - y)

$$CN^-(aq.) + H_2O(l) \rightleftharpoons HCN(aq.) + OH^-(aq.); \qquad K_{b(CN^-)} = K_h = \frac{K_w}{K_a}$$

At eqm. (x - y) – y y

$$K_{SP} = x\,(x - y) \qquad ...(i)$$

$$K_h = \frac{y^2}{(x-y)} \qquad ...(ii)$$

Points to remember

Solubility of AgCN decreases in basic solution [NaOH(aq.)/KOH(aq.)] due to common in effect.

Solubility of AgCN in acidic solution

Let solubility of AgCN is 'x' in acidic medium.

$$AgCN(s) \rightleftharpoons Ag^+(aq.) + CN^-(aq.) \quad ; \quad K_{SP}$$

At eqm. – x (x – y)

$$CN^-(aq.) + H^+(aq.) \rightleftharpoons HCN(aq.) \quad ; \quad K_{eq} = \frac{1}{K_{a(HCN)}} \cong 10^9$$

At eqm. (x – y) (z – y) y

$\simeq (z - x)$ $\simeq x$

$$AgCN(s) + H^+(aq.) \rightleftharpoons Ag^+(aq.) + HCN(aq.)$$

– $\simeq (z - x)$ x $\cong x$

Almost all CN^- is converted into HCN as the value of equilibrium constant is very high. ($y \simeq x$)

$$\boxed{K_{SP} \cdot \frac{1}{K_a} = \frac{x^2}{(z-x)}}$$; (z – x) = Concentration of H^+ at equilibrium.

$$y = x$$

$$\frac{K_{SP}}{K_a} = \frac{x^2}{[H^+]_{solution}}$$

where solubility is 'x' in acidic solution.

Note

Solubility of AgCN is maximum in acidic solution.

Solubility of AgCN:

In acidic medium > In water > In basic solution

Complex ion formation

We will learn this concept through a problem discussed below:

$$Cu^{2+}(aq.) + 4NH_3(aq.) \rightleftharpoons Cu(NH_3)_4^{2+}(aq.)$$

$K_{formation}$ or K_{form} or K_f $[Cu(NH_3)_4^{2+}]$ is equilibrium constant of above reaction.

$$Cu(NH_3)_4^{2+}(aq.) \rightleftharpoons Cu^{2+}(aq.) + 4NH_3(aq.);$$

For this reaction we write equilibrium constant as:

$$K_{Instability\ constant} \text{ or } K_{inst} \text{ or } K_d$$

$$\boxed{K_{form} = \frac{1}{K_{Inst.}}}$$

$$Ag^+(aq.) + 2NH_3(aq.) \rightleftharpoons Ag(NH_3)_2^+(aq.)$$

$$K_{formation} \text{ or } K_{form} = \frac{[Ag(NH_3)_2^+(aq.)]}{[Ag^+(aq.)][NH_3(aq.)]^2}$$

If constants of steps are provided then:

$$Ag^+(aq.) + NH_3(aq.) \rightleftharpoons Ag(NH_3)^+(aq.); \quad K_{f_1}$$

$$Ag(NH_3)^+(aq.) + NH_3(aq.) \rightleftharpoons Ag(NH_3)_2^+(aq.); \quad K_{f_2}$$

$$Ag^+(aq.) + 2NH_3(aq.) \rightleftharpoons Ag(NH_3)_2^+(aq.); \quad K_f = K_{f_1} K_{f_2}$$

$$Ag(NH_3)_2^+(aq.) \rightleftharpoons Ag^+(aq.) + 2NH_3(aq.)$$

$$K_d \text{ or } \boxed{K_{Inst.} = \frac{1}{K_f}}$$

Solubility of AgBr in aqueous solution of NH_3

Let solubility of AgBr is x

	$AgBr(s) \rightleftharpoons Ag^+(aq.) + Br^-(aq.);$			K_{sp}
At eqm.	–	$(x - y)$	x	
	$Ag^+(aq.) + 2NH_3(aq.) \rightleftharpoons Ag(NH_3)_2^+(aq.);$			K_f
At eqm.	$(x - y)$	$(z - 2y)$	y	
Overall	$AgBr(s) + 2NH_3(aq.) \rightleftharpoons Ag(NH_3)_2^+(aq.) + Br^-(aq.)$			
Rxn at eqm.	–	$(z - 2y)$	y	x

Due to very large value of K_f, we can assume $y = x$

$$K_{sp}.K_f = \frac{x^2}{(z - 2x)^2}$$